BB3

Technical User Manual

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Revision History

Version	Date	Author	Description
1.0	2026-07-06	Engineering Documentation Generator	Initial release

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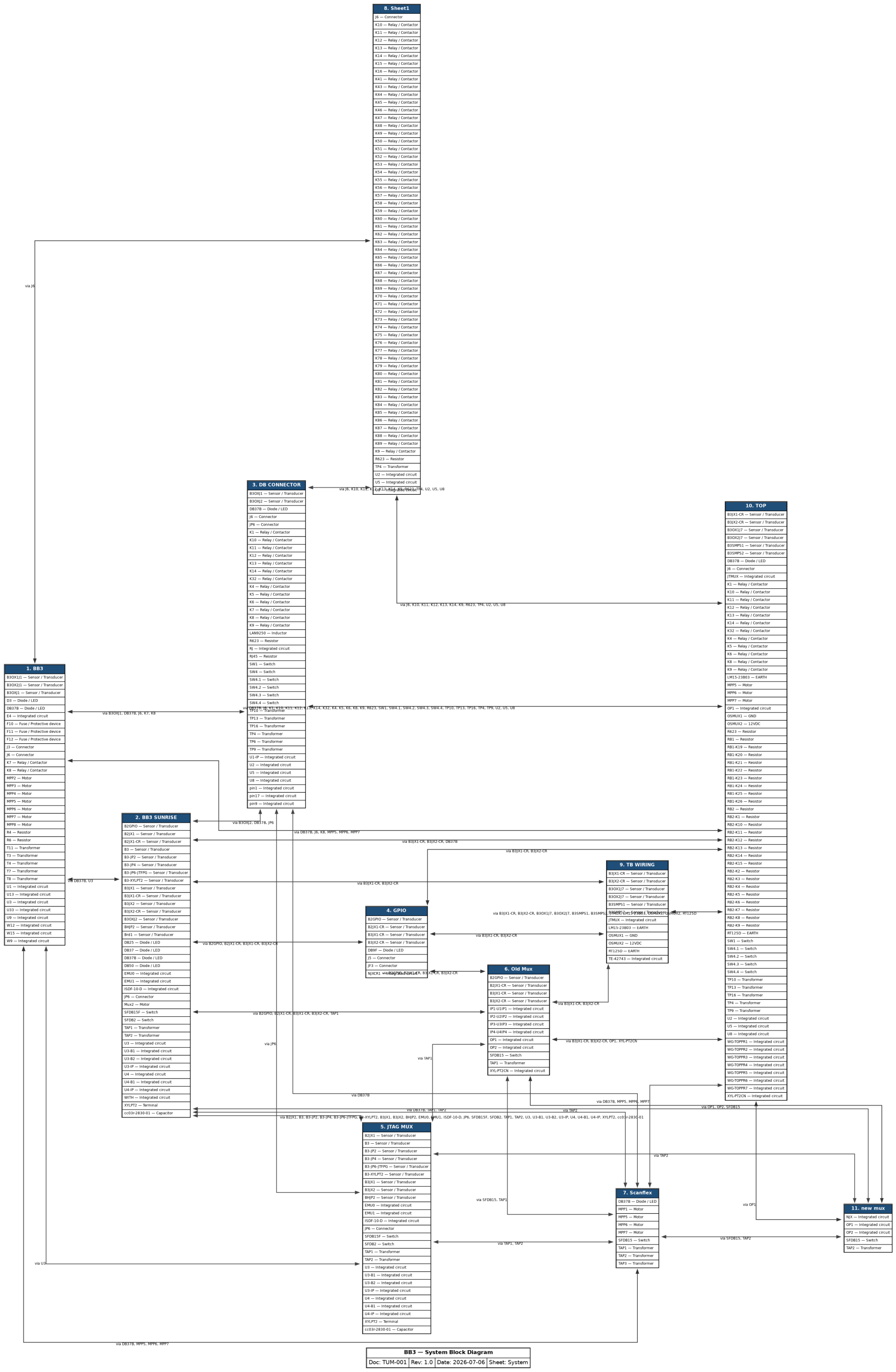
1. System Overview



The BB3 is organized into 11 functional subsystems: BB3, GPIO, DB CONNECTOR, BB3 SUNRISE, Sheet1, JTAG MUX, TB WIRING, TOP, Scanflex, Old Mux, new mux.

BB3: The BB3 sub-module interconnects 35 components of the system.

GPIO: The GPIO sub-module interconnects 8 components of the system.

DB CONNECTOR: The DB CONNECTOR sub-module interconnects 41 components of the system.

BB3 SUNRISE: The BB3 SUNRISE sub-module interconnects 38 components of the system.

Sheet1: The Sheet1 sub-module interconnects 62 components of the system.

JTAG MUX: The JTAG MUX sub-module interconnects 26 components of the system.

TB WIRING: The TB WIRING sub-module interconnects 12 components of the system.

TOP: The TOP sub-module interconnects 76 components of the system.

Scanflex: The Scanflex sub-module interconnects 9 components of the system.

Old Mux: The Old Mux sub-module interconnects 13 components of the system.

new mux: The new mux sub-module interconnects 5 components of the system.

Subsystem integration: B2GPIO spans BB3 SUNRISE and GPIO and Old Mux, carrying ground, other signals across the boundary. B2JX1 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B2JX1-CR spans BB3 SUNRISE and GPIO and Old Mux, carrying other signals across the boundary. B3 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3-JP2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3-JP4 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3-JP6-JTFPG spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3-XYLPT2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3JX1 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3JX1-CR spans BB3 SUNRISE and GPIO and Old Mux and TB WIRING and TOP, carrying ground, other signals across the boundary. B3JX2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. B3JX2-CR spans BB3 SUNRISE and GPIO and Old Mux and TB WIRING and TOP, carrying ground, other signals across the boundary. B3OX1J7 spans TB WIRING and TOP, carrying ground, other signals across the boundary. B3OX2J7 spans TB WIRING and TOP, carrying other signals across the boundary. B3OXJ1 spans BB3 and DB CONNECTOR, carrying ground, other signals across the boundary. B3OXJ2 spans BB3 SUNRISE and DB CONNECTOR, carrying other signals across the boundary. B3SMPS1 spans TB WIRING and TOP, carrying ground, other signals across the boundary. B3SMPS2 spans TB WIRING and TOP, carrying ground, other signals across the boundary. BHJP2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. DB37B spans BB3 and BB3 SUNRISE and DB CONNECTOR and Scanflex and TOP, carrying analog / sense, data / communication, other signals across the boundary. EMU0 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. EMU1 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. ISDF-10-D spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. J6 spans BB3 and DB CONNECTOR and Sheet1 and TOP, carrying analog / sense, data / communication, ground, other signals across the boundary. JP6 spans BB3 SUNRISE and DB CONNECTOR and JTAG MUX, carrying other signals across the boundary. JTMUX spans TB WIRING and TOP, carrying other signals across the boundary. K1 spans DB CONNECTOR and TOP, carrying other signals across the boundary. K10 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. K11 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. K12 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. K13 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. K14 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. K32 spans DB CONNECTOR and TOP, carrying other signals across the boundary. K4 spans DB CONNECTOR and TOP, carrying other signals across the boundary. K5 spans DB CONNECTOR and TOP, carrying other signals across the boundary. K6 spans DB CONNECTOR and TOP, carrying other signals across the boundary. K7 spans BB3 and DB CONNECTOR, carrying data / communication, other signals across the boundary. K8 spans BB3 and DB CONNECTOR and TOP, carrying data / communication, other signals across the boundary. K9 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. LM15-23B03 spans TB WIRING and TOP, carrying ground, other signals across the boundary. MPP5 spans BB3 and Scanflex and TOP, carrying other signals across the boundary. MPP6 spans BB3 and Scanflex and TOP, carrying data / communication, other signals across the boundary. MPP7 spans BB3 and Scanflex and TOP, carrying analog / sense, other signals across the boundary. OP1 spans Old Mux and TOP and new mux, carrying analog / sense, ground, other signals across the boundary. OP2 spans Old Mux and new mux, carrying other signals across the boundary. OSMUX1 spans TB WIRING and TOP, carrying ground, other signals across the boundary. OSMUX2 spans TB WIRING and TOP, carrying other signals across the boundary. R623 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. RT125D spans TB WIRING and TOP, carrying ground, other signals across the boundary. SFDB15 spans Old Mux and Scanflex and new mux, carrying other signals across the boundary. SFDB15F spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. SFDB2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. SW1 spans DB CONNECTOR and TOP, carrying other signals across the boundary. SW4.1 spans DB CONNECTOR and TOP, carrying other signals across the boundary. SW4.2 spans DB CONNECTOR and TOP, carrying other signals across the boundary. SW4.3 spans DB CONNECTOR and TOP, carrying other signals across the boundary. SW4.4 spans DB CONNECTOR and TOP, carrying other signals across the boundary. TAP1 spans BB3 SUNRISE and JTAG MUX and Old Mux and Scanflex, carrying other signals across the boundary. TAP2 spans BB3 SUNRISE and JTAG MUX and Scanflex and new mux, carrying other signals across the boundary. TP10 spans DB CONNECTOR and TOP, carrying other signals across the boundary. TP13 spans DB CONNECTOR and TOP, carrying other signals across the boundary. TP16 spans DB CONNECTOR and TOP, carrying other signals across the boundary. TP4 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. TP9 spans DB CONNECTOR and TOP, carrying other signals across the boundary. U2 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. U3 spans BB3 and BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U3-B1 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U3-B2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U3-IP spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U4 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U4-B1 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U4-IP spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. U5 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. U8 spans DB CONNECTOR and Sheet1 and TOP, carrying other signals across the boundary. XYL-PT2CN spans Old Mux and TOP, carrying analog / sense, ground, other signals across the boundary. XYLPT2 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary. cc03r-2830-01 spans BB3 SUNRISE and JTAG MUX, carrying other signals across the boundary.

Sub-Module Interconnections

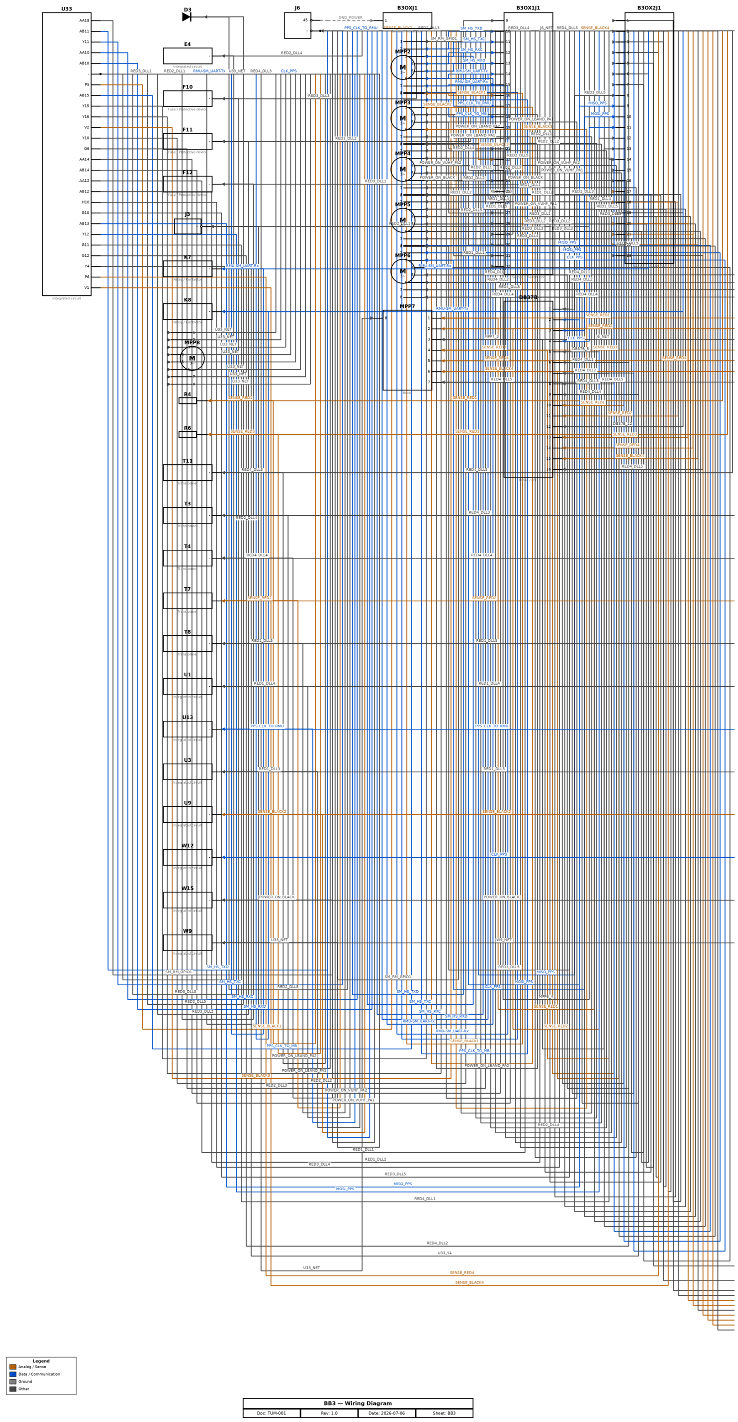
Bridging Component	Sub-Modules	Signal Classes
B2GPIO	BB3_SUNRISE / GPIO / Old_Mux	Ground, Other
B2JX1	BB3_SUNRISE / JTAG_MUX	Other
B2JX1-CR	BB3_SUNRISE / GPIO / Old_Mux	Other
B3	BB3_SUNRISE / JTAG_MUX	Other
B3-JP2	BB3_SUNRISE / JTAG_MUX	Other
B3-JP4	BB3_SUNRISE / JTAG_MUX	Other
B3-JP6-JTFPG	BB3_SUNRISE / JTAG_MUX	Other
B3-XYLPT2	BB3_SUNRISE / JTAG_MUX	Other
B3JX1	BB3_SUNRISE / JTAG_MUX	Other
B3JX1-CR	BB3_SUNRISE / GPIO / Old_Mux / TB_WIRING / TOP	Ground, Other

Bridging Component	Sub-Modules	Signal Classes
B3JX2	BB3_SUNRISE / JTAG_MUX	Other
B3JX2-CR	BB3_SUNRISE / GPIO / Old_Mux / TB_WIRING / TOP	Ground, Other
B3OX1J7	TB_WIRING / TOP	Ground, Other
B3OX2J7	TB_WIRING / TOP	Other
B3OXJ1	BB3 / DB_CONNECTOR	Ground, Other
B3OXJ2	BB3_SUNRISE / DB_CONNECTOR	Other
B3SMPS1	TB_WIRING / TOP	Ground, Other
B3SMPS2	TB_WIRING / TOP	Ground, Other
BHJP2	BB3_SUNRISE / JTAG_MUX	Other
DB37B	BB3 / BB3_SUNRISE / DB_CONNECTOR / Scanflex / TOP	Analog / Sense, Data / Communication, Other
EMU0	BB3_SUNRISE / JTAG_MUX	Other
EMU1	BB3_SUNRISE / JTAG_MUX	Other
ISDF-10-D	BB3_SUNRISE / JTAG_MUX	Other
J6	BB3 / DB_CONNECTOR / Sheet1 / TOP	Analog / Sense, Data / Communication, Ground, Other
JP6	BB3_SUNRISE / DB_CONNECTOR / JTAG_MUX	Other
JTMUX	TB_WIRING / TOP	Other
K1	DB_CONNECTOR / TOP	Other
K10	DB_CONNECTOR / Sheet1 / TOP	Other
K11	DB_CONNECTOR / Sheet1 / TOP	Other
K12	DB_CONNECTOR / Sheet1 / TOP	Other
K13	DB_CONNECTOR / Sheet1 / TOP	Other
K14	DB_CONNECTOR / Sheet1 / TOP	Other
K32	DB_CONNECTOR / TOP	Other
K4	DB_CONNECTOR / TOP	Other
K5	DB_CONNECTOR / TOP	Other
K6	DB_CONNECTOR / TOP	Other
K7	BB3 / DB_CONNECTOR	Data / Communication, Other
K8	BB3 / DB_CONNECTOR / TOP	Data / Communication, Other
K9	DB_CONNECTOR / Sheet1 / TOP	Other
LM15-23B03	TB_WIRING / TOP	Ground, Other
MPP5	BB3 / Scanflex / TOP	Other
MPP6	BB3 / Scanflex / TOP	Data / Communication, Other
MPP7	BB3 / Scanflex / TOP	Analog / Sense, Other
OP1	Old_Mux / TOP / new_mux	Analog / Sense, Ground, Other
OP2	Old_Mux / new_mux	Other
OSMUX1	TB_WIRING / TOP	Ground, Other
OSMUX2	TB_WIRING / TOP	Other
R623	DB_CONNECTOR / Sheet1 / TOP	Other
RT125D	TB_WIRING / TOP	Ground, Other

Bridging Component	Sub-Modules	Signal Classes
SFDB15	Old_Mux / Scanflex / new_mux	Other
SFDB15F	BB3_SUNRISE / JTAG_MUX	Other
SFDB2	BB3_SUNRISE / JTAG_MUX	Other
SW1	DB_CONNECTOR / TOP	Other
SW4.1	DB_CONNECTOR / TOP	Other
SW4.2	DB_CONNECTOR / TOP	Other
SW4.3	DB_CONNECTOR / TOP	Other
SW4.4	DB_CONNECTOR / TOP	Other
TAP1	BB3_SUNRISE / JTAG_MUX / Old_Mux / Scanflex	Other
TAP2	BB3_SUNRISE / JTAG_MUX / Scanflex / new_mux	Other
TP10	DB_CONNECTOR / TOP	Other
TP13	DB_CONNECTOR / TOP	Other
TP16	DB_CONNECTOR / TOP	Other
TP4	DB_CONNECTOR / Sheet1 / TOP	Other
TP9	DB_CONNECTOR / TOP	Other
U2	DB_CONNECTOR / Sheet1 / TOP	Other
U3	BB3 / BB3_SUNRISE / JTAG_MUX	Other
U3-B1	BB3_SUNRISE / JTAG_MUX	Other
U3-B2	BB3_SUNRISE / JTAG_MUX	Other
U3-IP	BB3_SUNRISE / JTAG_MUX	Other
U4	BB3_SUNRISE / JTAG_MUX	Other
U4-B1	BB3_SUNRISE / JTAG_MUX	Other
U4-IP	BB3_SUNRISE / JTAG_MUX	Other
U5	DB_CONNECTOR / Sheet1 / TOP	Other
U8	DB_CONNECTOR / Sheet1 / TOP	Other
XYL-PT2CN	Old_Mux / TOP	Analog / Sense, Ground, Other
XYLPT2	BB3_SUNRISE / JTAG_MUX	Other
cc03r-2830-01	BB3_SUNRISE / JTAG_MUX	Other

2. Sub-Module Descriptions

2.1 BB3



The BB3 sub-module interconnects 35 components of the system.

Digital communication between the stage's devices is carried over SM_HS_TXD, SM_HS_TXC, SM_HS_RXC, SM_HS_RXD, RMU-SM_UART-Tx, RMU-SM_UART-Rx, PPS_CLK_TO_RMU, PPS_CLK_TO_MB, MISO_PPS, MOSI_PPS, CLK_PPS. Operating conditions are monitored via SENSE_BLACK1, SENSE_BLACK2, SENSE_BLACK3, SENSE_RED1, SENSE_RED2, SENSE_RED3, SENSE_RED4, SENSE_BLACK4. A common ground reference is maintained throughout the stage.

The sub-module integrates 35 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the BB3 sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 SUNRISE sub-module through DB37B, U3 (signals: DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…). It interfaces with the DB CONNECTOR sub-module through B3OXJ1, DB37B, J6, K7, K8 (signals: B3RJSTL1_2, B3RJSTL2_1, B3RJSTL2_3, B3RJSTL4_6, B3RJSTL5_3, DB37B_1, DB37B_10, DB37B_11…). It interfaces with the JTAG MUX sub-module through U3 (signals: U3_IP, cc03r-2830-01_g). It interfaces with the Scanflex sub-module through DB37B, MPP5, MPP6, MPP7 (signals: MPP1_4, MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6…). It interfaces with the Sheet1 sub-module through J6 (signals: J6_100, J6_103, J6_104, J6_22, J6_24, J6_25, J6_26, J6_27…). It interfaces with the TOP sub-module through DB37B, J6, K8, MPP5, MPP6, MPP7 (signals: ETH_RXN, ETH_RXN-2, ETH_RXP, ETH_RXP-2, ETH_TXP, ETH_TXP-2, EtH_TXN, EtH_TXN-2…).

Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3 SUNRISE	DB37B, U3	DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…
DB CONNECTOR	B3OXJ1, DB37B, J6, K7, K8	B3RJSTL1_2, B3RJSTL2_1, B3RJSTL2_3, B3RJSTL4_6, B3RJSTL5_3, DB37B_1, DB37B_10, DB37B_11…
JTAG MUX	U3	U3_IP, cc03r-2830-01_g
Scanflex	DB37B, MPP5, MPP6, MPP7	MPP1_4, MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6…
Sheet1	J6	J6_100, J6_103, J6_104, J6_22, J6_24, J6_25, J6_26, J6_27…
TOP	DB37B, J6, K8, MPP5, MPP6, MPP7	ETH_RXN, ETH_RXN-2, ETH_RXP, ETH_RXP-2, ETH_TXP, ETH_TXP-2, EtH_TXN, EtH_TXN-2…

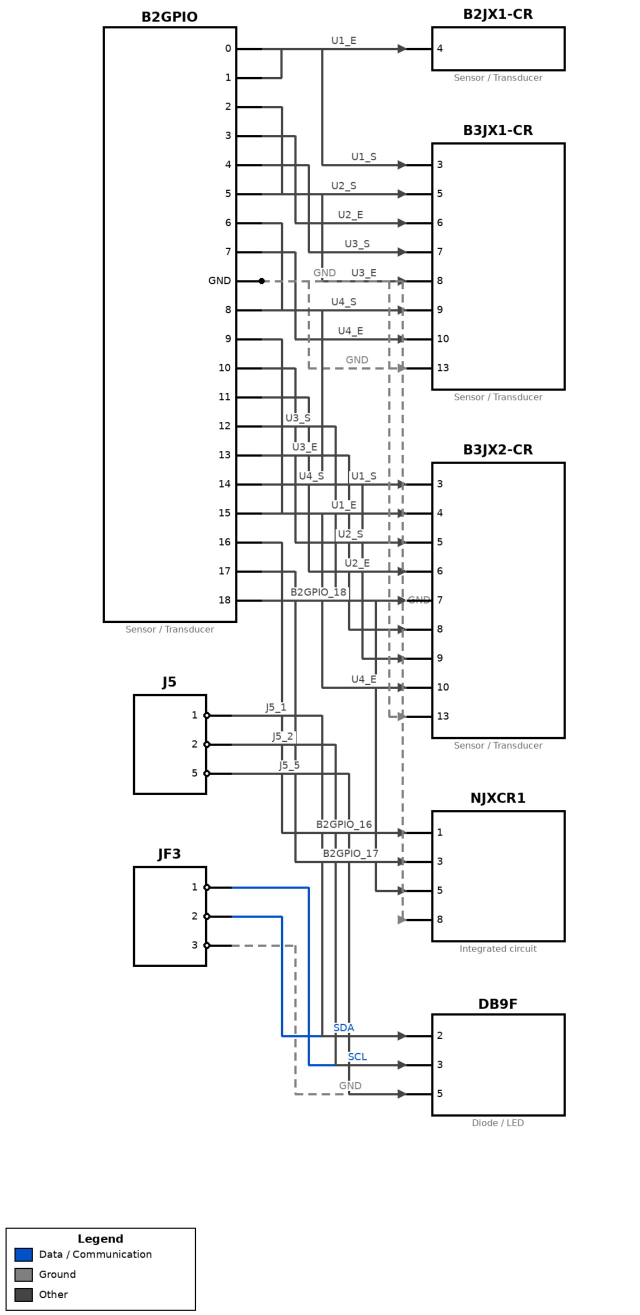
Signal List

Signal	Class	Connections
GND_POWER	Ground	J6.45 → B3OXJ1.1
CLK_PPS	Data / Communication	U33.- → W12.-; W12.- → J6.-; J6.- → MPP6.3; MPP6.3 → DB37B.4; DB37B.4 → B3OX2J1.11
MISO_PPS	Data / Communication	U33.AB13 → J6.-; J6.- → MPP6.1; MPP6.1 → DB37B.2; DB37B.2 → B3OX2J1.9
MOSI_PPS	Data / Communication	U33.Y12 → J6.-; J6.- → MPP6.2; MPP6.2 → DB37B.3; DB37B.3 → B3OX2J1.10
PPS_CLK_TO_MB	Data / Communication	U33.AB15 → J6.-; J6.- → MPP3.2; MPP3.2 → B3OX1J1.18
PPS_CLK_TO_RMU	Data / Communication	U33.- → U13.-; U13.- → J6.-; J6.- → MPP3.1; MPP3.1 → B3OX1J1.17
RMU-SM_UART-Rx	Data / Communication	U33.- → K7.-; K7.- → J6.-; J6.- → MPP2.7; MPP2.7 → B3OX1J1.15
RMU-SM_UART-Tx	Data / Communication	U33.- → K8.-; K8.- → J6.-; J6.- → MPP2.6; MPP2.6 → B3OX1J1.14
SM_HS_RXC	Data / Communication	U33.AA10 → J6.-; J6.- → MPP2.4; MPP2.4 → B3OX1J1.12

Signal	Class	Connections
SM_HS_RXD	Data / Communication	U33.AB10 → J6.-; J6.- → MPP2.5; MPP2.5 → B3OX1J1.13
SM_HS_TXC	Data / Communication	U33.Y11 → J6.-; J6.- → MPP2.3; MPP2.3 → B3OX1J1.11
SM_HS_TXD	Data / Communication	U33.AB11 → J6.-; J6.- → MPP2.2; MPP2.2 → B3OX1J1.10
SENSE_BLACK1	Analog / Sense	U33.P5 → J6.-; J6.- → MPP2.8; MPP2.8 → B3OX1J1.16
SENSE_BLACK2	Analog / Sense	U33.- → U9.-; U9.- → J6.-; J6.- → MPP3.3; MPP3.3 → B3OX1J1.19
SENSE_BLACK3	Analog / Sense	U33.V2 → J6.-; J6.- → MPP3.6; MPP3.6 → B3OX1J1.22
SENSE_BLACK4	Analog / Sense	U33.V1 → J6.-; J6.- → MPP7.6; MPP7.6 → DB37B.15; DB37B.15 → B3OX2J1.22
SENSE_RED1	Analog / Sense	U33.- → R4.-; R4.- → J6.-; J6.- → MPP7.1; MPP7.1 → DB37B.10; DB37B.10 → B3OX2J1.17
SENSE_RED2	Analog / Sense	U33.- → T7.-; T7.- → J6.-; J6.- → MPP7.2; MPP7.2 → DB37B.11; DB37B.11 → B3OX2J1.18
SENSE_RED3	Analog / Sense	U33.- → R6.-; R6.- → J6.-; J6.- → MPP7.4; MPP7.4 → DB37B.13; DB37B.13 → B3OX2J1.20
SENSE_RED4	Analog / Sense	U33.P6 → J6.-; J6.- → MPP7.5; MPP7.5 → DB37B.14; DB37B.14 → B3OX2J1.21
DB37B_12	Other	DB37B.12 → B3OX2J1.19
DB37B_5	Other	DB37B.5 → B3OX2J1.12
J6_NET	Other	J6.- → MPP6.4; J6.- → MPP7.3
MPP6_4	Other	MPP6.4 → DB37B.5
MPP7_3	Other	MPP7.3 → DB37B.12
POWER_ON_BLACK	Other	U33.- → W15.-; W15.- → J6.-; J6.- → MPP4.6; MPP4.6 → B3OX1J1.30
POWER_ON_LBAND_PA1	Other	U33.Y16 → J6.-; J6.- → MPP3.5; MPP3.5 → B3OX1J1.21
POWER_ON_LBAND_PA2	Other	U33.Y15 → J6.-; J6.- → MPP3.4; MPP3.4 → B3OX1J1.20
POWER_ON_VUHF_PA1	Other	U33.AB14 → J6.-; J6.- → MPP4.5; MPP4.5 → B3OX1J1.29
POWER_ON_VUHF_PA2	Other	U33.AA14 → J6.-; J6.- → MPP4.4; MPP4.4 → B3OX1J1.28
RED1_DLL1	Other	U33.AA12 → J6.-; J6.- → MPP4.7; MPP4.7 → B3OX1J1.31
RED1_DLL2	Other	U33.AB12 → J6.-; J6.- → MPP4.8; MPP4.8 → B3OX1J1.32
RED1_DLL3	Other	U33.- → U3.-; U3.- → J6.-; J6.- → MPP5.1; MPP5.1 → B3OX2J1.1
RED1_DLL4	Other	U33.- → U1.-; U1.- → J6.-; J6.- → MPP5.2; MPP5.2 → B3OX2J1.2
RED1_DLL5	Other	U33.- → T8.-; T8.- → J6.-; J6.- → MPP5.3; MPP5.3 → B3OX2J1.3
RED2_DLL1	Other	U33.- → J3.-; J3.- → J6.-; J6.- → MPP3.7; MPP3.7 → B3OX1J1.23
RED2_DLL2	Other	U33.Y10 → J6.-; J6.- → MPP3.8; MPP3.8 → B3OX1J1.24
RED2_DLL3	Other	U33.G6 → J6.-; J6.- → MPP4.1; MPP4.1 → B3OX1J1.25
RED2_DLL4	Other	U33.- → E4.-; E4.- → J6.-; J6.- → MPP4.2; MPP4.2 → B3OX1J1.26
RED2_DLL5	Other	U33.- → D3.-; D3.- → J6.-; J6.- → MPP4.3; MPP4.3 → B3OX1J1.27
RED3_DLL1	Other	U33.- → F12.-; F12.- → J6.-; J6.- → MPP5.4; MPP5.4 → B3OX2J1.4
RED3_DLL2	Other	U33.- → F11.-; F11.- → J6.-; J6.- → MPP5.5; MPP5.5 → B3OX2J1.5
RED3_DLL3	Other	U33.- → F10.-; F10.- → J6.-; J6.- → MPP5.6; MPP5.6 → B3OX2J1.6
RED3_DLL4	Other	U33.H10 → J6.-; J6.- → MPP5.7; MPP5.7 → B3OX2J1.7
RED3_DLL5	Other	U33.G10 → J6.-; J6.- → MPP5.8; MPP5.8 → DB37B.1; DB37B.1 → B3OX2J1.8
RED4_DLL1	Other	U33.G11 → J6.-; J6.- → MPP6.5; MPP6.5 → DB37B.6; DB37B.6 → B3OX2J1.13

Signal	Class	Connections
RED4_DLL2	Other	U33.G12 → J6.-; J6.- → MPP6.6; MPP6.6 → DB37B.7; DB37B.7 → B3OX2J1.14
RED4_DLL3	Other	U33.- → T3.-; T3.- → J6.-; J6.- → MPP6.7; MPP6.7 → DB37B.8; DB37B.8 → B3OX2J1.15
RED4_DLL4	Other	U33.- → T4.-; T4.- → J6.-; J6.- → MPP6.8; MPP6.8 → DB37B.9; DB37B.9 → B3OX2J1.16
RED4_DLL5	Other	U33.- → T11.-; T11.- → J6.-; J6.- → MPP7.7; MPP7.7 → DB37B.16; DB37B.16 → B3OX2J1.23
SM_RM_GPIO1	Other	U33.AA18 → J6.-; J6.- → MPP2.1; MPP2.1 → B3OX1J1.9
U33_NET	Other	U33.- → W9.-; U33.- → MPP7.8; U33.- → MPP8.1; U33.- → MPP8.2; U33.- → MPP8.3; U33.- → MPP8.4; U33.- → MPP8.5; U33.- → MPP8.6; U33.- → MPP8.7; U33.- → MPP8.8
U33_Y4	Other	U33.Y4 → J6.-
W9_NET	Other	W9.- → J6.-

2.2 GPIO



GPIO — Wiring Diagram			
Doc: TUM-001	Rev: 1.0	Date: 2026-07-06	Sheet: GPIO

The GPIO sub-module interconnects 8 components of the system.

Digital communication between the stage's devices is carried over SCL, SDA. A common ground reference is maintained throughout the stage.

The sub-module integrates 8 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the GPIO sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 SUNRISE sub-module through B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR (signals: GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…). It interfaces with the Old Mux sub-module through B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR (signals: B2JX1-CR_4, B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8…). It interfaces with the TB WIRING sub-module through B3JX1-CR, B3JX2-CR (signals: JTMUX_1, JTMUX_2). It interfaces with the TOP sub-module through B3JX1-CR, B3JX2-CR (signals: JTMUX_1, JTMUX_2).

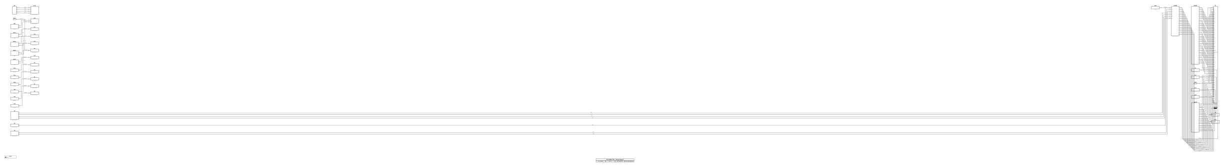
Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3 SUNRISE	B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR	GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…
Old Mux	B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR	B2JX1-CR_4, B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8…
TB WIRING	B3JX1-CR, B3JX2-CR	JTMUX_1, JTMUX_2
TOP	B3JX1-CR, B3JX2-CR	JTMUX_1, JTMUX_2

Signal List

Signal	Class	Connections
GND	Ground	JF3.3 → DB9F.5; B2GPIO.GND → B3JX1-CR.13; B2GPIO.GND → B3JX2-CR.13; B2GPIO.GND → NJXCR1.8
SCL	Data / Communication	JF3.1 → DB9F.3
SDA	Data / Communication	JF3.2 → DB9F.2
B2GPIO_16	Other	B2GPIO.16 → NJXCR1.1
B2GPIO_17	Other	B2GPIO.17 → NJXCR1.3
B2GPIO_18	Other	B2GPIO.18 → NJXCR1.5
J5_1	Other	J5.1 → DB9F.2
J5_2	Other	J5.2 → DB9F.3
J5_5	Other	J5.5 → DB9F.5
U1_E	Other	B2GPIO.1 → B2JX1-CR.4; B2GPIO.9 → B3JX2-CR.4
U1_S	Other	B2GPIO.0 → B3JX1-CR.3; B2GPIO.8 → B3JX2-CR.3
U2_E	Other	B2GPIO.3 → B3JX1-CR.6; B2GPIO.11 → B3JX2-CR.6
U2_S	Other	B2GPIO.2 → B3JX1-CR.5; B2GPIO.10 → B3JX2-CR.5
U3_E	Other	B2GPIO.5 → B3JX1-CR.8; B2GPIO.13 → B3JX2-CR.8
U3_S	Other	B2GPIO.4 → B3JX1-CR.7; B2GPIO.12 → B3JX2-CR.7
U4_E	Other	B2GPIO.7 → B3JX1-CR.10; B2GPIO.15 → B3JX2-CR.10
U4_S	Other	B2GPIO.6 → B3JX1-CR.9; B2GPIO.14 → B3JX2-CR.9

2.3 DB CONNECTOR



The DB CONNECTOR sub-module interconnects 41 components of the system.

The sub-module integrates 41 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the DB CONNECTOR sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 sub-module through B3OXJ1, DB37B, J6, K7, K8 (signals: CLK_PPS, DB37B_12, DB37B_5, GND_POWER, J6_NET, MISO_PPS, MOSI_PPS, MPP6_4…). It interfaces with the BB3 SUNRISE sub-module through B3OXJ2, DB37B, JP6 (signals: DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…). It interfaces with the JTAG MUX sub-module through JP6 (signals: JP6_NET). It interfaces with the Scanflex sub-module through DB37B (signals: MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6, MPP6_7…). It interfaces with the Sheet1 sub-module through J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8 (signals: J6_100, J6_103, J6_104, J6_22, J6_24, J6_25, J6_26, J6_27…). It interfaces with the TOP sub-module through DB37B, J6, K1, K10, K11, K12, K13, K14, K32, K4, K5, K6, K8, K9, R623, SW1, SW4.1, SW4.2, SW4.3, SW4.4, TP10, TP13, TP16, TP4, TP9, U2, U5, U8 (signals: ETH_RXN, ETH_RXN-2, ETH_RXP, ETH_RXP-2, ETH_TXP, ETH_TXP-2, EtH_TXN, EtH_TXN-2…).

Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3	B3OXJ1, DB37B, J6, K7, K8	CLK_PPS, DB37B_12, DB37B_5, GND_POWER, J6_NET, MISO_PPS, MOSI_PPS, MPP6_4…
BB3 SUNRISE	B3OXJ2, DB37B, JP6	DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…
JTAG MUX	JP6	JP6_NET
Scanflex	DB37B	MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6, MPP6_7…
Sheet1	J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8	J6_100, J6_103, J6_104, J6_22, J6_24, J6_25, J6_26, J6_27…
TOP	DB37B, J6, K1, K10, K11, K12, K13, K14, K32, K4, K5, K6, K8, K9, R623, SW1, SW4.1, SW4.2, SW4.3, SW4.4, TP10, TP13, TP16, TP4, TP9, U2, U5, U8	ETH_RXN, ETH_RXN-2, ETH_RXP, ETH_RXP-2, ETH_TXP, ETH_TXP-2, EtH_TXN, EtH_TXN-2…

Signal List

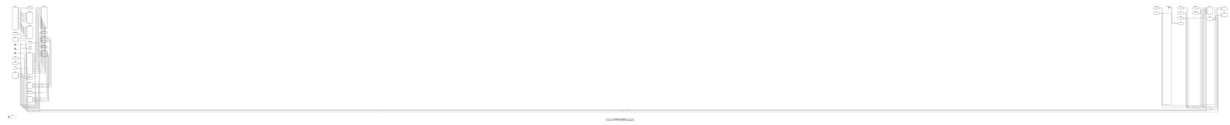
Signal	Class	Connections
B3OXJ2_17	Other	B3OXJ2.17 → SW4.-
B3RJSTL1_2	Other	J6.2 → RJ45.-
B3RJSTL2_1	Other	J6.18 → pin1.-; pin1.- → pin17.-

Signal	Class	Connections
B3RJSTL2_3	Other	J6.19 → pin9.-; pin9.- → J6.-
B3RJSTL4_6	Other	J6.98 → RJ45.-
B3RJSTL5_3	Other	J6.11 → RJ45.-
DB37B_1	Other	DB37B.1 → B3OXJ2.1
DB37B_10	Other	DB37B.10 → B3OXJ2.10
DB37B_11	Other	DB37B.11 → B3OXJ2.11
DB37B_12	Other	DB37B.12 → B3OXJ2.12
DB37B_13	Other	DB37B.13 → B3OXJ2.13
DB37B_14	Other	DB37B.14 → B3OXJ2.14
DB37B_15	Other	DB37B.15 → B3OXJ2.15
DB37B_16	Other	DB37B.16 → B3OXJ2.16
DB37B_2	Other	DB37B.2 → B3OXJ2.2
DB37B_3	Other	DB37B.3 → B3OXJ2.3
DB37B_4	Other	DB37B.4 → B3OXJ2.4
DB37B_5	Other	DB37B.5 → B3OXJ2.5
DB37B_6	Other	DB37B.6 → B3OXJ2.6
DB37B_7	Other	DB37B.7 → B3OXJ2.7
DB37B_8	Other	DB37B.8 → B3OXJ2.8
DB37B_9	Other	DB37B.9 → B3OXJ2.9
J6_100	Other	J6.100 → DB37B.14
J6_103	Other	J6.103 → DB37B.15
J6_104	Other	J6.104 → DB37B.16
J6_22	Other	J6.22 → B3OXJ1.2
J6_24	Other	J6.24 → B3OXJ1.3
J6_25	Other	J6.25 → B3OXJ1.4
J6_26	Other	J6.26 → B3OXJ1.5
J6_27	Other	J6.27 → B3OXJ1.6
J6_36	Other	J6.36 → B3OXJ1.7
J6_37	Other	J6.37 → B3OXJ1.8
J6_39	Other	J6.39 → B3OXJ1.9
J6_40	Other	J6.40 → B3OXJ1.10
J6_41	Other	J6.41 → B3OXJ1.11
J6_42	Other	J6.42 → B3OXJ1.12
J6_43	Other	J6.43 → B3OXJ1.13
J6_44	Other	J6.44 → B3OXJ1.14
J6_45	Other	J6.45 → B3OXJ1.1
J6_46	Other	J6.46 → B3OXJ1.15
J6_47	Other	J6.47 → B3OXJ1.16
J6_48	Other	J6.48 → B3OXJ1.17
J6_49	Other	J6.49 → B3OXJ1.18
J6_50	Other	J6.50 → B3OXJ1.19
J6_51	Other	J6.51 → B3OXJ1.20

Signal	Class	Connections
J6_53	Other	J6.53 → B3OXJ1.21
J6_54	Other	J6.54 → B3OXJ1.22
J6_55	Other	J6.55 → B3OXJ1.23
J6_57	Other	J6.57 → B3OXJ1.24
J6_58	Other	J6.58 → B3OXJ1.25
J6_59	Other	J6.59 → B3OXJ1.26
J6_60	Other	J6.60 → B3OXJ1.27
J6_61	Other	J6.61 → B3OXJ1.28
J6_62	Other	J6.62 → B3OXJ1.29
J6_63	Other	J6.63 → B3OXJ1.30
J6_64	Other	J6.64 → B3OXJ1.31
J6_65	Other	J6.65 → B3OXJ1.32
J6_66	Other	J6.66 → DB37B.1
J6_77	Other	J6.77 → DB37B.2
J6_78	Other	J6.78 → DB37B.3
J6_79	Other	J6.79 → DB37B.4
J6_80	Other	J6.80 → DB37B.5
J6_83	Other	J6.83 → DB37B.6
J6_84	Other	J6.84 → DB37B.7
J6_85	Other	J6.85 → DB37B.8
J6_86	Other	J6.86 → DB37B.9
J6_88	Other	J6.88 → DB37B.10
J6_89	Other	J6.89 → DB37B.11
J6_90	Other	J6.90 → DB37B.12
J6_99	Other	J6.99 → DB37B.13
J6_NET	Other	J6.- → K1.-; J6.- → K32.-
JP6_10	Other	JP6.10 → U1-IP.3
JP6_4	Other	JP6.4 → U1-IP.5
JP6_6	Other	JP6.6 → U1-IP.4
JP6_8	Other	JP6.8 → U1-IP.6
K1_NET	Other	K1.- → LAN9250.-
K32_NET	Other	K32.- → RJ.45
R623_NET	Other	R623.- → K9.-
SW1_1	Other	SW1.1 → K10.C
SW1_2	Other	SW1.2 → K10.NO
SW4.1_1	Other	SW4.1.1 → K11.-
SW4.1_2	Other	SW4.1.2 → K11.-
SW4.2_1	Other	SW4.2.1 → K12.-
SW4.2_2	Other	SW4.2.2 → K12.-
SW4.3_1	Other	SW4.3.1 → K13.-
SW4.3_2	Other	SW4.3.2 → K13.-
SW4.4_1	Other	SW4.4.1 → K14.-

Signal	Class	Connections
SW4.4_2	Other	SW4.4.2 → K14.-
TP10_NET	Other	TP10.- → K4.-
TP13_NET	Other	TP13.- → K6.-
TP16_NET	Other	TP16.- → K5.-
TP4_NET	Other	TP4.- → K9.-
TP6_NET	Other	TP6.- → K7.-
TP9_NET	Other	TP9.- → K8.-
U2_2	Other	U2.2 → B3OXJ2.20
U2_3	Other	U2.3 → B3OXJ2.21
U2_4	Other	U2.4 → B3OXJ2.22
U2_5	Other	U2.5 → B3OXJ2.23
U5_3	Other	U5.3 → B3OXJ2.17
U8_4	Other	U8.4 → B3OXJ2.18
U8_5	Other	U8.5 → B3OXJ2.19

2.4 BB3 SUNRISE



The BB3 SUNRISE sub-module interconnects 38 components of the system.

A common ground reference is maintained throughout the stage.

The sub-module integrates 38 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the BB3 SUNRISE sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 sub-module through DB37B, U3 (signals: CLK_PPS, DB37B_12, DB37B_5, MISO_PPS, MOSI_PPS, MPP6_4, MPP7_3, RED1_DLL3…). It interfaces with the DB CONNECTOR sub-module through B3OXJ2, DB37B, JP6 (signals: B3OXJ2_17, DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15…). It interfaces with the GPIO sub-module through B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR (signals: B2GPIO_16, B2GPIO_17, B2GPIO_18, GND, U1_E, U1_S, U2_E, U2_S…). It interfaces with the JTAG MUX sub-module through B2JX1, B3, B3-JP2, B3-JP4, B3-JP6-JTFPG, B3-XYLPT2, B3JX1, B3JX2, BHJP2, EMU0, EMU1, ISDF-10-D, JP6, SFDB15F, SFDB2, TAP1, TAP2, U3, U3-B1, U3-B2, U3-IP, U4, U4-B1, U4-IP, XYLPT2, cc03r-2830-01 (signals: B3-JP2_JX1, B3-JP4_JX1, B3-JP6-JTFPG_NET, B3-XYLPT2_NET, B3JX2_IP3, B3JX2_OP5, B3JX2_OP7, B3_TAP1…). It interfaces with the Old Mux sub-module through B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR, TAP1 (signals: B2JX1-CR_4, B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8…). It interfaces with the Scanflex sub-module through DB37B, TAP1, TAP2 (signals: MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6, MPP6_7…). It interfaces with the TB WIRING sub-module through B3JX1-CR, B3JX2-CR (signals: JTMUX_1, JTMUX_2). It interfaces with the TOP sub-module through B3JX1-CR, B3JX2-CR, DB37B (signals: JTMUX_1, JTMUX_2, MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5…). It interfaces with the new mux sub-module through TAP2 (signals: TAP2_NET).

Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3	DB37B, U3	CLK_PPS, DB37B_12, DB37B_5, MISO_PPS, MOSI_PPS, MPP6_4, MPP7_3, RED1_DLL3…
DB CONNECTOR	B3OXJ2, DB37B, JP6	B3OXJ2_17, DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15…
GPIO	B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR	B2GPIO_16, B2GPIO_17, B2GPIO_18, GND, U1_E, U1_S, U2_E, U2_S…
JTAG MUX	B2JX1, B3, B3-JP2, B3-JP4, B3-JP6-JTFPG, B3-XYLPT2, B3JX1, B3JX2, BHJP2, EMU0, EMU1, ISDF-10-D, JP6, SFDB15F, SFDB2, TAP1, TAP2, U3, U3-B1, U3-B2, U3-IP, U4, U4-B1, U4-IP, XYLPT2, cc03r-2830-01	B3-JP2_JX1, B3-JP4_JX1, B3-JP6-JTFPG_NET, B3-XYLPT2_NET, B3JX2_IP3, B3JX2_OP5, B3JX2_OP7, B3_TAP1…
Old Mux	B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR, TAP1	B2JX1-CR_4, B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8…

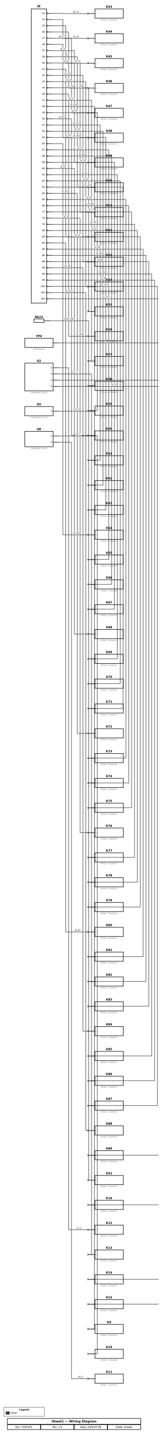
Connected Sub-Module	Via Components	Signals
Scanflex	DB37B, TAP1, TAP2	MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6, MPP6_7…
TB WIRING	B3JX1-CR, B3JX2-CR	JTMUX_1, JTMUX_2
TOP	B3JX1-CR, B3JX2-CR, DB37B	JTMUX_1, JTMUX_2, MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5…
new mux	TAP2	TAP2_NET

Signal List

Signal	Class	Connections
GND	Ground	B2GPIO.GND → B3JX1-CR.13; B2GPIO.GND → B3JX2-CR.13
B3-JP2_JX1	Other	B3-JP2.JX1 → ISDF-10-D.M
B3-JP4_JX1	Other	B3-JP4.JX1 → B3JX2.IP4
B3-JP6-JTFPG_NET	Other	B3-JP6-JTFPG.- → B2JX1.IP2
B3-XYLPT2_NET	Other	B3-XYLPT2.- → XYLPT2.-
B3JX2_IP3	Other	B3JX2.IP3 → TAP2.-
B3JX2_OP5	Other	B3JX2.OP5 → B3-JP2.JX1
B3JX2_OP7	Other	B3JX2.OP7 → B3-JP4.JX1
B3_TAP1	Other	B3.TAP1 → SFDB15F.-
B3_TAP2	Other	B3.TAP2 → SFDB2.-
BHJP2_A	Other	BHJP2.A → B3JX2.OP5
BHJP2_B	Other	BHJP2.B → B3JX2.OP7
DB25_NET	Other	DB25.- → WITH.75; DB25.- → Mux2.-
DB37B_1	Other	DB37B.1 → B3OXJ2.1
DB37B_10	Other	DB37B.10 → B3OXJ2.10
DB37B_11	Other	DB37B.11 → B3OXJ2.11
DB37B_12	Other	DB37B.12 → B3OXJ2.12
DB37B_13	Other	DB37B.13 → B3OXJ2.13
DB37B_14	Other	DB37B.14 → B3OXJ2.14
DB37B_15	Other	DB37B.15 → B3OXJ2.15
DB37B_16	Other	DB37B.16 → B3OXJ2.16
DB37B_2	Other	DB37B.2 → B3OXJ2.2
DB37B_3	Other	DB37B.3 → B3OXJ2.3
DB37B_4	Other	DB37B.4 → B3OXJ2.4
DB37B_5	Other	DB37B.5 → B3OXJ2.5
DB37B_6	Other	DB37B.6 → B3OXJ2.6
DB37B_7	Other	DB37B.7 → B3OXJ2.7
DB37B_8	Other	DB37B.8 → B3OXJ2.8
DB37B_9	Other	DB37B.9 → B3OXJ2.9
DB37_NET	Other	DB37.- → DB37B.1
DB50_NET	Other	DB50.- → WITH.75
EMU0_NET	Other	EMU0.- → U4-B1.3
EMU1_NET	Other	EMU1.- → U4-B1.4

Signal	Class	Connections
ISDF-10-D_M	Other	ISDF-10-D.M → cc03r-2830-01.g
JP6_NET	Other	JP6.- → TAP1.-; JP6.- → B3-JP6-JTFPG.-
SFDB15F_NET	Other	SFDB15F.- → TAP1.JP1
SFDB2_NET	Other	SFDB2.- → B3JX2.OP6
TAP1_JP1	Other	TAP1.JP1 → B3JX1.OP2
TAP1_NET	Other	TAP1.- → B3.TAP1
TAP2_NET	Other	TAP2.- → B3.TAP2
TCK	Other	U3-B1.6 → U3-IP.6; U3-IP.6 → U3-B2.6
TDI	Other	U3-B1.5 → U3-IP.5; U3-IP.5 → U3-B2.5
TDO	Other	U3-B1.4 → U3-IP.4; U3-IP.4 → U3-B2.4
TMS	Other	U3-B1.3 → U3-IP.3; U3-IP.3 → U3-B2.3
U1_E	Other	B2GPIO.1 → B2JX1-CR.4; B2GPIO.9 → B3JX2-CR.4
U1_S	Other	B2GPIO.0 → B3JX1-CR.3; B2GPIO.8 → B3JX2-CR.3
U2_E	Other	B2GPIO.3 → B3JX1-CR.6; B2GPIO.11 → B3JX2-CR.6
U2_S	Other	B2GPIO.2 → B3JX1-CR.5; B2GPIO.10 → B3JX2-CR.5
U3_E	Other	B2GPIO.5 → B3JX1-CR.8; B2GPIO.13 → B3JX2-CR.8
U3_IP	Other	U3.IP → U4.IP
U3_S	Other	B2GPIO.4 → B3JX1-CR.7; B2GPIO.12 → B3JX2-CR.7
U4-B1_2	Other	U4-B1.2 → U4-IP.3
U4-B1_3	Other	U4-B1.3 → U4-IP.4
U4-B1_4	Other	U4-B1.4 → U4-IP.5
U4-B1_5	Other	U4-B1.5 → U4-IP.6
U4_E	Other	B2GPIO.7 → B3JX1-CR.10; B2GPIO.15 → B3JX2-CR.10
U4_IP	Other	U4.IP → B3JX2.IP3
U4_S	Other	B2GPIO.6 → B3JX1-CR.9; B2GPIO.14 → B3JX2-CR.9
WITH_75	Other	WITH.75 → Brd1.-
XYLPT2_NET	Other	XYLPT2.- → B3JX1.OP1
cc03r-2830-01_g	Other	cc03r-2830-01.g → U3.IP

2.5 Sheet1



The Sheet1 sub-module interconnects 62 components of the system.

The sub-module integrates 62 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the Sheet1 sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 sub-module through J6 (signals: CLK_PPS, GND_POWER, J6_NET, MISO_PPS, MOSI_PPS, POWER_ON_BLACK, POWER_ON_LBAND_PA1, POWER_ON_LBAND_PA2…). It interfaces with the DB CONNECTOR sub-module through J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8 (signals: B3RJSTL1_2, B3RJSTL2_1, B3RJSTL2_3, B3RJSTL4_6, B3RJSTL5_3, J6_100, J6_103, J6_104…). It interfaces with the TOP sub-module through J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8 (signals: ETH_RXN, ETH_RXN-2, ETH_RXP, ETH_RXP-2, ETH_TXP, ETH_TXP-2, EtH_TXN, EtH_TXN-2…).

Interconnections with Other Sub-Modules

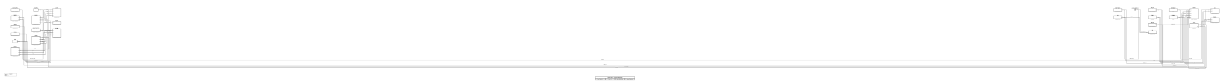
Connected Sub-Module	Via Components	Signals
BB3	J6	CLK_PPS, GND_POWER, J6_NET, MISO_PPS, MOSI_PPS, POWER_ON_BLACK, POWER_ON_LBAND_PA1, POWER_ON_LBAND_PA2…
DB CONNECTOR	J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8	B3RJSTL1_2, B3RJSTL2_1, B3RJSTL2_3, B3RJSTL4_6, B3RJSTL5_3, J6_100, J6_103, J6_104…
TOP	J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8	ETH_RXN, ETH_RXN-2, ETH_RXP, ETH_RXP-2, ETH_TXP, ETH_TXP-2, EtH_TXN, EtH_TXN-2…

Signal List

Signal	Class	Connections
J6_100	Other	J6.100 → K87.-
J6_103	Other	J6.103 → K88.-
J6_104	Other	J6.104 → K89.-
J6_22	Other	J6.22 → K43.-
J6_24	Other	J6.24 → K44.-
J6_25	Other	J6.25 → K45.-
J6_26	Other	J6.26 → K46.-
J6_27	Other	J6.27 → K47.-
J6_36	Other	J6.36 → K48.-
J6_37	Other	J6.37 → K49.-
J6_39	Other	J6.39 → K50.-
J6_40	Other	J6.40 → K51.-
J6_41	Other	J6.41 → K52.-
J6_42	Other	J6.42 → K53.-
J6_43	Other	J6.43 → K54.-
J6_44	Other	J6.44 → K55.-
J6_46	Other	J6.46 → K56.-
J6_47	Other	J6.47 → K57.-
J6_48	Other	J6.48 → K58.-
J6_49	Other	J6.49 → K59.-
J6_50	Other	J6.50 → K60.-
J6_51	Other	J6.51 → K61.-

Signal	Class	Connections
J6_53	Other	J6.53 → K62.-
J6_54	Other	J6.54 → K63.-
J6_55	Other	J6.55 → K64.-
J6_57	Other	J6.57 → K65.-
J6_58	Other	J6.58 → K66.-
J6_59	Other	J6.59 → K67.-
J6_60	Other	J6.60 → K68.-
J6_61	Other	J6.61 → K69.-
J6_62	Other	J6.62 → K70.-
J6_63	Other	J6.63 → K71.-
J6_64	Other	J6.64 → K72.-
J6_65	Other	J6.65 → K73.-
J6_66	Other	J6.66 → K74.-
J6_77	Other	J6.77 → K75.-
J6_78	Other	J6.78 → K76.-
J6_79	Other	J6.79 → K77.-
J6_80	Other	J6.80 → K78.-
J6_83	Other	J6.83 → K79.-
J6_84	Other	J6.84 → K80.-
J6_85	Other	J6.85 → K81.-
J6_86	Other	J6.86 → K82.-
J6_88	Other	J6.88 → K83.-
J6_89	Other	J6.89 → K84.-
J6_90	Other	J6.90 → K85.-
J6_99	Other	J6.99 → K86.-
R623_NET	Other	R623.- → K41.-
TP4_NET	Other	TP4.- → K16.-
U2_2	Other	U2.2 → K12.-
U2_3	Other	U2.3 → K13.-
U2_4	Other	U2.4 → K14.-
U2_5	Other	U2.5 → K15.-
U5_3	Other	U5.3 → K9.-
U8_4	Other	U8.4 → K10.-
U8_5	Other	U8.5 → K11.-

2.6 JTAG MUX



The JTAG MUX sub-module interconnects 26 components of the system.

The sub-module integrates 26 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the JTAG MUX sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 sub-module through U3 (signals: RED1_DLL3). It interfaces with the BB3 SUNRISE sub-module through B2JX1, B3, B3-JP2, B3-JP4, B3-JP6-JTFPG, B3-XYLPT2, B3JX1, B3JX2, BHJP2, EMU0, EMU1, ISDF-10-D, JP6, SFDB15F, SFDB2, TAP1, TAP2, U3, U3-B1, U3-B2, U3-IP, U4, U4-B1, U4-IP, XYLPT2, cc03r-2830-01 (signals: B3-JP2_JX1, B3-JP4_JX1, B3-JP6-JTFPG_NET, B3-XYLPT2_NET, B3JX2_IP3, B3JX2_OP5, B3JX2_OP7, B3_TAP1…). It interfaces with the DB CONNECTOR sub-module through JP6 (signals: JP6_10, JP6_4, JP6_6, JP6_8). It interfaces with the Old Mux sub-module through TAP1 (signals: OP2_U1B2, TAP1_NET). It interfaces with the Scanflex sub-module through TAP1, TAP2 (signals: TAP1_NET, TAP2_NET). It interfaces with the new mux sub-module through TAP2 (signals: TAP2_NET).

Interconnections with Other Sub-Modules

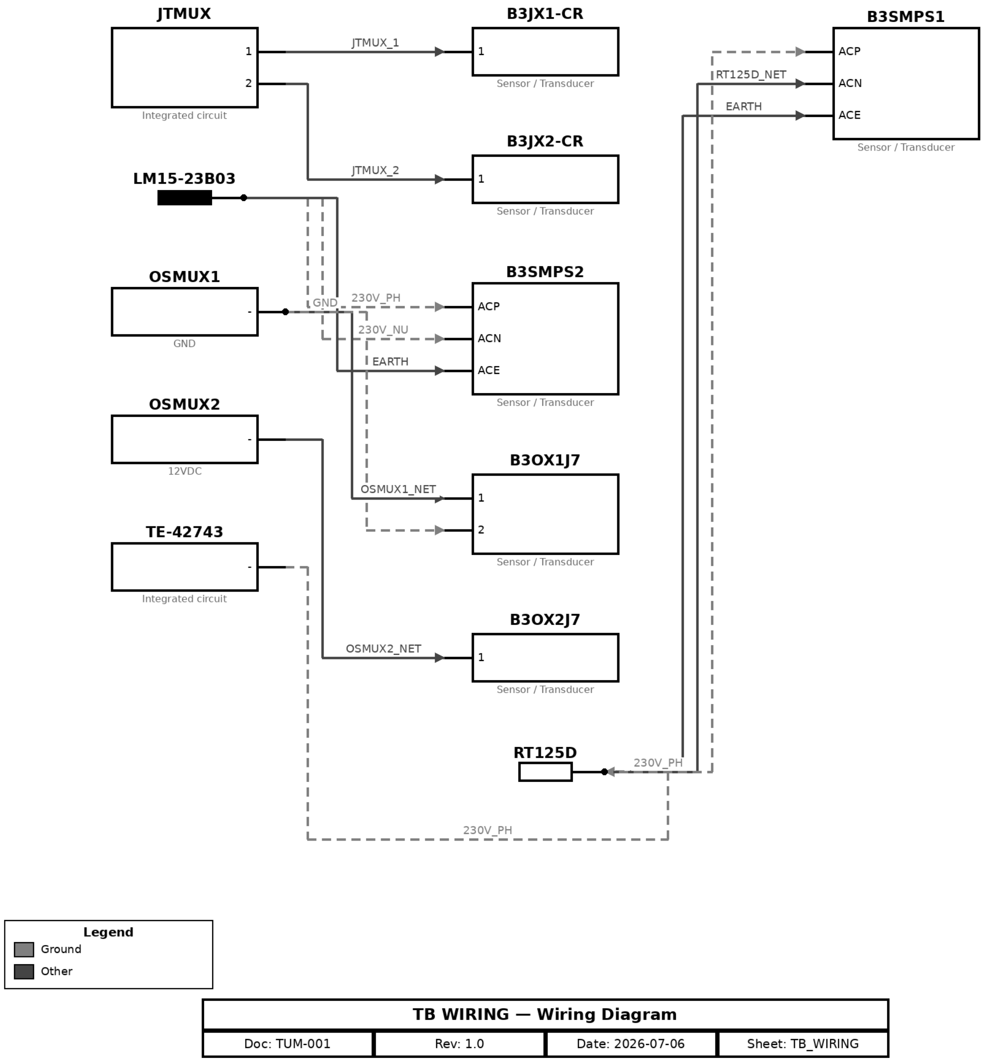
Connected Sub-Module	Via Components	Signals
BB3	U3	RED1_DLL3
BB3 SUNRISE	B2JX1, B3, B3-JP2, B3-JP4, B3-JP6-JTFPG, B3-XYLPT2, B3JX1, B3JX2, BHJP2, EMU0, EMU1, ISDF-10-D, JP6, SFDB15F, SFDB2, TAP1, TAP2, U3, U3-B1, U3-B2, U3-IP, U4, U4-B1, U4-IP, XYLPT2, cc03r-2830-01	B3-JP2_JX1, B3-JP4_JX1, B3-JP6-JTFPG_NET, B3-XYLPT2_NET, B3JX2_IP3, B3JX2_OP5, B3JX2_OP7, B3_TAP1…
DB CONNECTOR	JP6	JP6_10, JP6_4, JP6_6, JP6_8
Old Mux	TAP1	OP2_U1B2, TAP1_NET
Scanflex	TAP1, TAP2	TAP1_NET, TAP2_NET
new mux	TAP2	TAP2_NET

Signal List

Signal	Class	Connections
B3-JP2_JX1	Other	B3-JP2.JX1 → ISDF-10-D.M
B3-JP4_JX1	Other	B3-JP4.JX1 → B3JX2.IP4
B3-JP6-JTFPG_NET	Other	B3-JP6-JTFPG.- → B2JX1.IP2
B3-XYLPT2_NET	Other	B3-XYLPT2.- → XYLPT2.-
B3JX2_IP3	Other	B3JX2.IP3 → TAP2.-
B3JX2_OP5	Other	B3JX2.OP5 → B3-JP2.JX1
B3JX2_OP7	Other	B3JX2.OP7 → B3-JP4.JX1
B3_TAP1	Other	B3.TAP1 → SFDB15F.-
B3_TAP2	Other	B3.TAP2 → SFDB2.-

Signal	Class	Connections
BHJP2_A	Other	BHJP2.A → B3JX2.OP5
BHJP2_B	Other	BHJP2.B → B3JX2.OP7
EMU0_NET	Other	EMU0.- → U4-B1.3
EMU1_NET	Other	EMU1.- → U4-B1.4
ISDF-10-D_M	Other	ISDF-10-D.M → cc03r-2830-01.g
JP6_NET	Other	JP6.- → TAP1.-; JP6.- → B3-JP6-JTFPG.-
SFDB15F_NET	Other	SFDB15F.- → TAP1.JP1
SFDB2_NET	Other	SFDB2.- → B3JX2.OP6
TAP1_JP1	Other	TAP1.JP1 → B3JX1.OP2
TAP1_NET	Other	TAP1.- → B3.TAP1
TAP2_NET	Other	TAP2.- → B3.TAP2
TCK	Other	U3-B1.6 → U3-IP.6; U3-IP.6 → U3-B2.6
TDI	Other	U3-B1.5 → U3-IP.5; U3-IP.5 → U3-B2.5
TDO	Other	U3-B1.4 → U3-IP.4; U3-IP.4 → U3-B2.4
TMS	Other	U3-B1.3 → U3-IP.3; U3-IP.3 → U3-B2.3
U3_IP	Other	U3.IP → U4.IP
U4-B1_2	Other	U4-B1.2 → U4-IP.3
U4-B1_3	Other	U4-B1.3 → U4-IP.4
U4-B1_4	Other	U4-B1.4 → U4-IP.5
U4-B1_5	Other	U4-B1.5 → U4-IP.6
U4_IP	Other	U4.IP → B3JX2.IP3
XYLPT2_NET	Other	XYLPT2.- → B3JX1.OP1
cc03r-2830-01_g	Other	cc03r-2830-01.g → U3.IP

2.7 TB WIRING



The TB WIRING sub-module interconnects 12 components of the system.

A common ground reference is maintained throughout the stage.

The sub-module integrates 12 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the TB WIRING sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 SUNRISE sub-module through B3JX1-CR, B3JX2-CR (signals: GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…). It interfaces with the GPIO sub-module through B3JX1-CR, B3JX2-CR (signals: GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…). It interfaces with the Old Mux sub-module through B3JX1-CR, B3JX2-CR (signals: B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8, B3JX1-CR_9…). It interfaces with the TOP sub-module through B3JX1-CR, B3JX2-CR, B3OX1J7, B3OX2J7, B3SMPS1, B3SMPS2, JTMUX, LM15-23B03, OSMUX1, OSMUX2, RT125D (signals: 230V_NU, 230V_PH, EARTH, GND, JTMUX_1, JTMUX_2, OSMUX1_NET, OSMUX2_NET…).

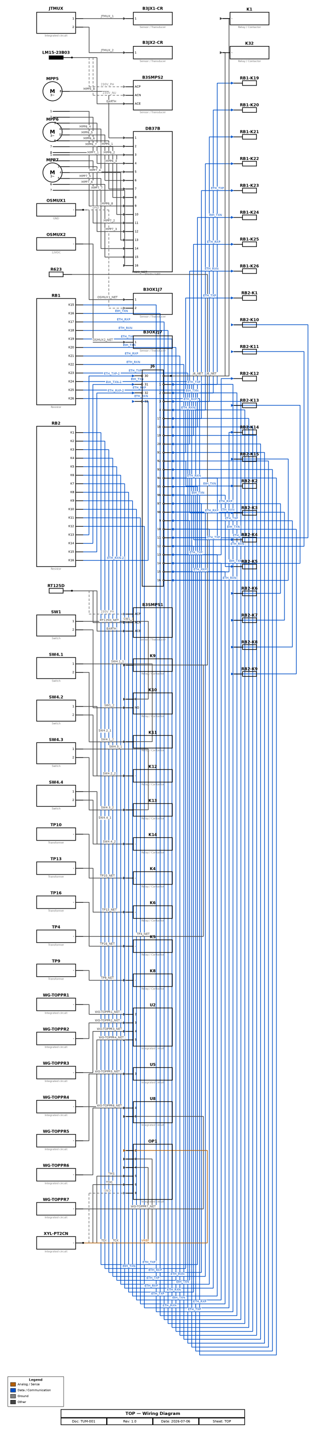
Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3 SUNRISE	B3JX1-CR, B3JX2-CR	GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…
GPIO	B3JX1-CR, B3JX2-CR	GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…
Old Mux	B3JX1-CR, B3JX2-CR	B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8, B3JX1-CR_9…
TOP	B3JX1-CR, B3JX2-CR, B3OX1J7, B3OX2J7, B3SMPS1, B3SMPS2, JTMUX, LM15-23B03, OSMUX1, OSMUX2, RT125D	230V_NU, 230V_PH, EARTH, GND, JTMUX_1, JTMUX_2, OSMUX1_NET, OSMUX2_NET…

Signal List

Signal	Class	Connections
230V_NU	Ground	LM15-23B03.- → B3SMPS2.ACN
230V_PH	Ground	TE-42743.- → RT125D.-; RT125D.- → B3SMPS1.ACP; LM15-23B03.- → B3SMPS2.ACP
GND	Ground	OSMUX1.- → B3OX1J7.2
EARTH	Other	RT125D.- → B3SMPS1.ACE; LM15-23B03.- → B3SMPS2.ACE
JTMUX_1	Other	JTMUX.1 → B3JX1-CR.1
JTMUX_2	Other	JTMUX.2 → B3JX2-CR.1
OSMUX1_NET	Other	OSMUX1.- → B3OX1J7.1
OSMUX2_NET	Other	OSMUX2.- → B3OX2J7.1
RT125D_NET	Other	RT125D.- → B3SMPS1.ACN

2.8 TOP



The TOP sub-module interconnects 76 components of the system.

Digital communication between the stage's devices is carried over ETH_TXP, EtH_TXN, ETH_RXP, ETH_RXN, ETH_TXP-2, EtH_TXN-2, ETH_RXP-2, ETH_RXN-2. Operating conditions are monitored via VREF. A common ground reference is maintained throughout the stage.

The sub-module integrates 76 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the TOP sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 sub-module through DB37B, J6, K8, MPP5, MPP6, MPP7 (signals: CLK_PPS, DB37B_12, DB37B_5, GND_POWER, J6_NET, MISO_PPS, MOSI_PPS, MPP6_4…). It interfaces with the BB3 SUNRISE sub-module through B3JX1-CR, B3JX2-CR, DB37B (signals: DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…). It interfaces with the DB CONNECTOR sub-module through DB37B, J6, K1, K10, K11, K12, K13, K14, K32, K4, K5, K6, K8, K9, R623, SW1, SW4.1, SW4.2, SW4.3, SW4.4, TP10, TP13, TP16, TP4, TP9, U2, U5, U8 (signals: B3RJSTL1_2, B3RJSTL2_1, B3RJSTL2_3, B3RJSTL4_6, B3RJSTL5_3, DB37B_1, DB37B_10, DB37B_11…). It interfaces with the GPIO sub-module through B3JX1-CR, B3JX2-CR (signals: GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…). It interfaces with the Old Mux sub-module through B3JX1-CR, B3JX2-CR, OP1, XYL-PT2CN (signals: B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8, B3JX1-CR_9…). It interfaces with the Scanflex sub-module through DB37B, MPP5, MPP6, MPP7 (signals: MPP1_4, MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6…). It interfaces with the Sheet1 sub-module through J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8 (signals: J6_100, J6_103, J6_104, J6_22, J6_24, J6_25, J6_26, J6_27…). It interfaces with the TB WIRING sub-module through B3JX1-CR, B3JX2-CR, B3OX1J7, B3OX2J7, B3SMPS1, B3SMPS2, JTMUX, LM15-23B03, OSMUX1, OSMUX2, RT125D (signals: 230V_NU, 230V_PH, EARTH, GND, JTMUX_1, JTMUX_2, OSMUX1_NET, OSMUX2_NET…). It interfaces with the new mux sub-module through OP1 (signals: OP1_U1B2).

Interconnections with Other Sub-Modules

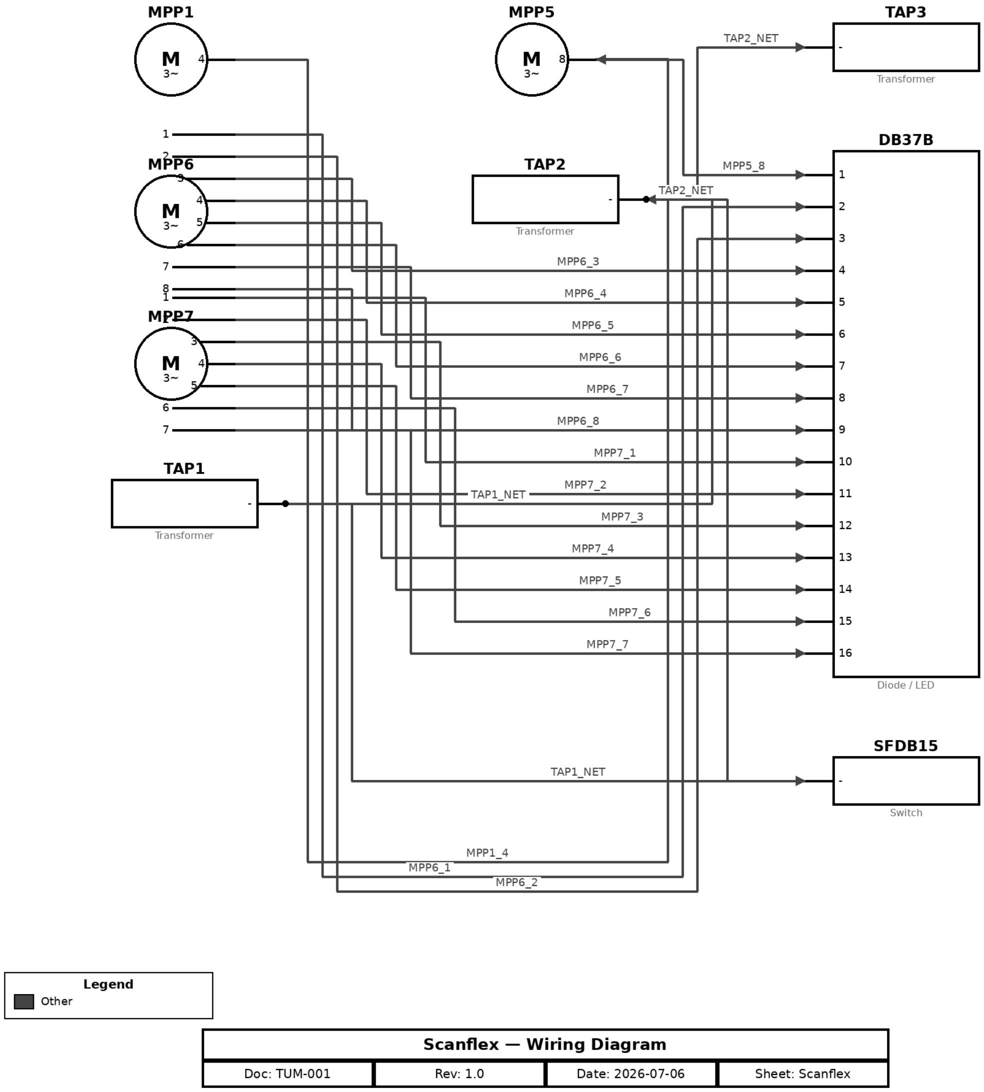
Connected Sub-Module	Via Components	Signals
BB3	DB37B, J6, K8, MPP5, MPP6, MPP7	CLK_PPS, DB37B_12, DB37B_5, GND_POWER, J6_NET, MISO_PPS, MOSI_PPS, MPP6_4…
BB3 SUNRISE	B3JX1-CR, B3JX2-CR, DB37B	DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…
DB CONNECTOR	DB37B, J6, K1, K10, K11, K12, K13, K14, K32, K4, K5, K6, K8, K9, R623, SW1, SW4.1, SW4.2, SW4.3, SW4.4, TP10, TP13, TP16, TP4, TP9, U2, U5, U8	B3RJSTL1_2, B3RJSTL2_1, B3RJSTL2_3, B3RJSTL4_6, B3RJSTL5_3, DB37B_1, DB37B_10, DB37B_11…
GPIO	B3JX1-CR, B3JX2-CR	GND, U1_E, U1_S, U2_E, U2_S, U3_E, U3_S, U4_E…
Old Mux	B3JX1-CR, B3JX2-CR, OP1, XYL-PT2CN	B3JX1-CR_10, B3JX1-CR_13, B3JX1-CR_3, B3JX1-CR_5, B3JX1-CR_6, B3JX1-CR_7, B3JX1-CR_8, B3JX1-CR_9…
Scanflex	DB37B, MPP5, MPP6, MPP7	MPP1_4, MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6…
Sheet1	J6, K10, K11, K12, K13, K14, K9, R623, TP4, U2, U5, U8	J6_100, J6_103, J6_104, J6_22, J6_24, J6_25, J6_26, J6_27…
TB WIRING	B3JX1-CR, B3JX2-CR, B3OX1J7, B3OX2J7, B3SMPS1, B3SMPS2, JTMUX, LM15-23B03, OSMUX1, OSMUX2, RT125D	230V_NU, 230V_PH, EARTH, GND, JTMUX_1, JTMUX_2, OSMUX1_NET, OSMUX2_NET…
new mux	OP1	OP1_U1B2

Signal List

Signal	Class	Connections
230V_NU	Ground	LM15-23B03.- → B3SMPS2.ACN
230V_PH	Ground	RT125D.- → B3SMPS1.ACP; LM15-23B03.- → B3SMPS2.ACP
GND	Ground	XYL-PT2CN.- → OP1.8; OSMUX1.- → B3OX1J7.2
ETH_RXN	Data / Communication	RB1.K18 → J6.4; J6.4 → RB1-K22.C; RB1.K22 → J6.20; J6.20 → RB1-K26.C; RB1.K26 → J6.94; J6.94 → RB2-K4.C; RB2.K4 → J6.98; J6.98 → RB2-K7.C; RB2.K8 → J6.12; J6.12 → RB2-K11.C; RB2.K12 → J6.16; J6.16 → RB2-K15.C
ETH_RXN-2	Data / Communication	RB2.K16 → J6.33
ETH_RXP	Data / Communication	RB1.K17 → J6.3; J6.3 → RB1-K21.C; RB1.K21 → J6.19; J6.19 → RB1-K25.C; RB1.K25 → J6.93; J6.93 → RB2-K3.C; RB2.K3 → J6.97; J6.97 → RB2-K6.C; RB2.K7 → J6.11; J6.11 → RB2-K10.C; RB2.K11 → J6.15; J6.15 → RB2-K14.C
ETH_RXP-2	Data / Communication	RB2.K15 → J6.32
ETH_TXP	Data / Communication	RB1.K15 → J6.1; J6.1 → RB1-K19.C; RB1.K19 → J6.17; J6.17 → RB1-K23.C; RB1.K23 → J6.91; J6.91 → RB2-K1.C; RB2.K1 → J6.95; J6.95 → RB2-K4.C; RB2.K5 → J6.9; J6.9 → RB2-K8.C; RB2.K9 → J6.13; J6.13 → RB2-K12.C
ETH_TXP-2	Data / Communication	RB2.K13 → J6.30
EtH_TXN	Data / Communication	RB1.K16 → J6.2; J6.2 → RB1-K20.C; RB1.K20 → J6.18; J6.18 → RB1-K24.C; RB1.K24 → J6.92; J6.92 → RB2-K2.C; RB2.K2 → J6.96; J6.96 → RB2-K5.C; RB2.K6 → J6.10; J6.10 → RB2-K9.C; RB2.K10 → J6.14; J6.14 → RB2-K13.C
EtH_TXN-2	Data / Communication	RB2.K14 → J6.31
VREF	Analog / Sense	XYL-PT2CN.- → OP1.2
EARTH	Other	RT125D.- → B3SMPS1.ACE; LM15-23B03.- → B3SMPS2.ACE
J6_NET	Other	J6.- → K1.-; J6.- → K32.-
JTMUX_1	Other	JTMUX.1 → B3JX1-CR.1
JTMUX_2	Other	JTMUX.2 → B3JX2-CR.1
MPP5_8	Other	MPP5.8 → DB37B.1
MPP6_1	Other	MPP6.1 → DB37B.2
MPP6_2	Other	MPP6.2 → DB37B.3
MPP6_3	Other	MPP6.3 → DB37B.4
MPP6_4	Other	MPP6.4 → DB37B.5
MPP6_5	Other	MPP6.5 → DB37B.6
MPP6_6	Other	MPP6.6 → DB37B.7
MPP6_7	Other	MPP6.7 → DB37B.8
MPP6_8	Other	MPP6.8 → DB37B.9
MPP7_1	Other	MPP7.1 → DB37B.10
MPP7_2	Other	MPP7.2 → DB37B.11
MPP7_3	Other	MPP7.3 → DB37B.12
MPP7_4	Other	MPP7.4 → DB37B.13
MPP7_5	Other	MPP7.5 → DB37B.14
MPP7_6	Other	MPP7.6 → DB37B.15
MPP7_7	Other	MPP7.7 → DB37B.16
OSMUX1_NET	Other	OSMUX1.- → B3OX1J7.1
OSMUX2_NET	Other	OSMUX2.- → B3OX2J7.1
R623_NET	Other	R623.- → K9.-
RT125D_NET	Other	RT125D.- → B3SMPS1.ACN
SW1_1	Other	SW1.1 → K10.C

Signal	Class	Connections
SW1_2	Other	SW1.2 → K10.NO
SW4.1_1	Other	SW4.1.1 → K11.-
SW4.1_2	Other	SW4.1.2 → K11.-
SW4.2_1	Other	SW4.2.1 → K12.-
SW4.2_2	Other	SW4.2.2 → K12.-
SW4.3_1	Other	SW4.3.1 → K13.-
SW4.3_2	Other	SW4.3.2 → K13.-
SW4.4_1	Other	SW4.4.1 → K14.-
SW4.4_2	Other	SW4.4.2 → K14.-
TCK	Other	XYL-PT2CN.- → OP1.4
TDI	Other	XYL-PT2CN.- → OP1.3
TDO	Other	XYL-PT2CN.- → OP1.6
TMS	Other	XYL-PT2CN.- → OP1.5
TP10_NET	Other	TP10.- → K4.-
TP13_NET	Other	TP13.- → K6.-
TP16_NET	Other	TP16.- → K5.-
TP4_NET	Other	TP4.- → K9.-
TP9_NET	Other	TP9.- → K8.-
WG-TOPPR1_NET	Other	WG-TOPPR1.- → U2.2
WG-TOPPR2_NET	Other	WG-TOPPR2.- → U2.3
WG-TOPPR3_NET	Other	WG-TOPPR3.- → U2.4
WG-TOPPR4_NET	Other	WG-TOPPR4.- → U2.5
WG-TOPPR5_NET	Other	WG-TOPPR5.- → U5.3
WG-TOPPR6_NET	Other	WG-TOPPR6.- → U8.4
WG-TOPPR7_NET	Other	WG-TOPPR7.- → U8.5

2.9 Scanflex



The Scanflex sub-module interconnects 9 components of the system.

The sub-module integrates 9 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the Scanflex sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 sub-module through DB37B, MPP5, MPP6, MPP7 (signals: CLK_PPS, DB37B_12, DB37B_5, J6_NET, MISO_PPS, MOSI_PPS, MPP6_4, MPP7_3…). It interfaces with the BB3 SUNRISE sub-module through DB37B, TAP1, TAP2 (signals: B3JX2_IP3, DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15…). It interfaces with the DB CONNECTOR sub-module through DB37B (signals: DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…). It interfaces with the JTAG MUX sub-module through TAP1, TAP2 (signals: B3JX2_IP3, JP6_NET, SFDB15F_NET, TAP1_JP1, TAP1_NET, TAP2_NET). It interfaces with the Old Mux sub-module through SFDB15, TAP1 (signals: OP2_U1B2, TAP1_NET). It interfaces with the TOP sub-module through DB37B, MPP5, MPP6, MPP7 (signals: MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6, MPP6_7…). It interfaces with the new mux sub-module through SFDB15, TAP2 (signals: SFDB15_NET, TAP2_NET).

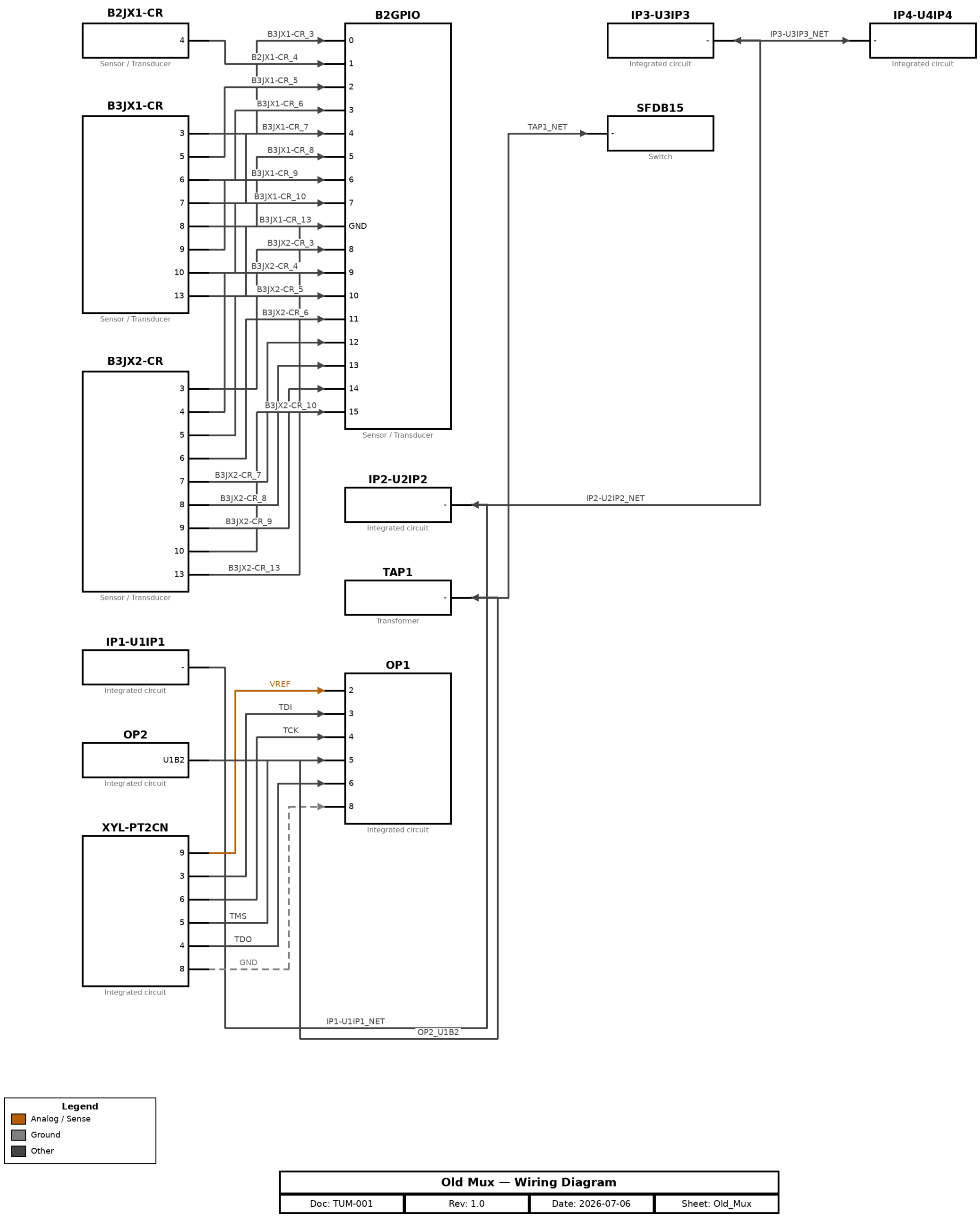
Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3	DB37B, MPP5, MPP6, MPP7	CLK_PPS, DB37B_12, DB37B_5, J6_NET, MISO_PPS, MOSI_PPS, MPP6_4, MPP7_3…
BB3 SUNRISE	DB37B, TAP1, TAP2	B3JX2_IP3, DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15…
DB CONNECTOR	DB37B	DB37B_1, DB37B_10, DB37B_11, DB37B_12, DB37B_13, DB37B_14, DB37B_15, DB37B_16…
JTAG MUX	TAP1, TAP2	B3JX2_IP3, JP6_NET, SFDB15F_NET, TAP1_JP1, TAP1_NET, TAP2_NET
Old Mux	SFDB15, TAP1	OP2_U1B2, TAP1_NET
TOP	DB37B, MPP5, MPP6, MPP7	MPP5_8, MPP6_1, MPP6_2, MPP6_3, MPP6_4, MPP6_5, MPP6_6, MPP6_7…
new mux	SFDB15, TAP2	SFDB15_NET, TAP2_NET

Signal List

Signal	Class	Connections
MPP1_4	Other	MPP1.4 → MPP5.8
MPP5_8	Other	MPP5.8 → DB37B.1
MPP6_1	Other	MPP6.1 → DB37B.2
MPP6_2	Other	MPP6.2 → DB37B.3
MPP6_3	Other	MPP6.3 → DB37B.4
MPP6_4	Other	MPP6.4 → DB37B.5
MPP6_5	Other	MPP6.5 → DB37B.6
MPP6_6	Other	MPP6.6 → DB37B.7
MPP6_7	Other	MPP6.7 → DB37B.8
MPP6_8	Other	MPP6.8 → DB37B.9
MPP7_1	Other	MPP7.1 → DB37B.10
MPP7_2	Other	MPP7.2 → DB37B.11
MPP7_3	Other	MPP7.3 → DB37B.12
MPP7_4	Other	MPP7.4 → DB37B.13
MPP7_5	Other	MPP7.5 → DB37B.14
MPP7_6	Other	MPP7.6 → DB37B.15
MPP7_7	Other	MPP7.7 → DB37B.16
TAP1_NET	Other	TAP1.- → TAP2.-; TAP1.- → SFDB15.-
TAP2_NET	Other	TAP2.- → TAP3.-; TAP2.- → SFDB15.-

2.10 Old Mux



The Old Mux sub-module interconnects 13 components of the system.

Operating conditions are monitored via VREF. A common ground reference is maintained throughout the stage.

The sub-module integrates 13 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the Old Mux sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 SUNRISE sub-module through B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR, TAP1 (signals: GND, JP6_NET, SFDB15F_NET, TAP1_JP1, TAP1_NET, U1_E, U1_S, U2_E…). It interfaces with the GPIO sub-module through B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR (signals: B2GPIO_16, B2GPIO_17, B2GPIO_18, GND, U1_E, U1_S, U2_E, U2_S…). It interfaces with the JTAG MUX sub-module through TAP1 (signals: JP6_NET, SFDB15F_NET, TAP1_JP1, TAP1_NET). It interfaces with the Scanflex sub-module through SFDB15, TAP1 (signals: TAP1_NET, TAP2_NET). It interfaces with the TB WIRING sub-module through B3JX1-CR, B3JX2-CR (signals: JTMUX_1, JTMUX_2). It interfaces with the TOP sub-module through B3JX1-CR, B3JX2-CR, OP1, XYL-PT2CN (signals: GND, JTMUX_1, JTMUX_2, TCK, TDI, TDO, TMS, VREF). It interfaces with the new mux sub-module through OP1, OP2, SFDB15 (signals: OP1_U1B2, SFDB15_NET, TAP2_NET).

Interconnections with Other Sub-Modules

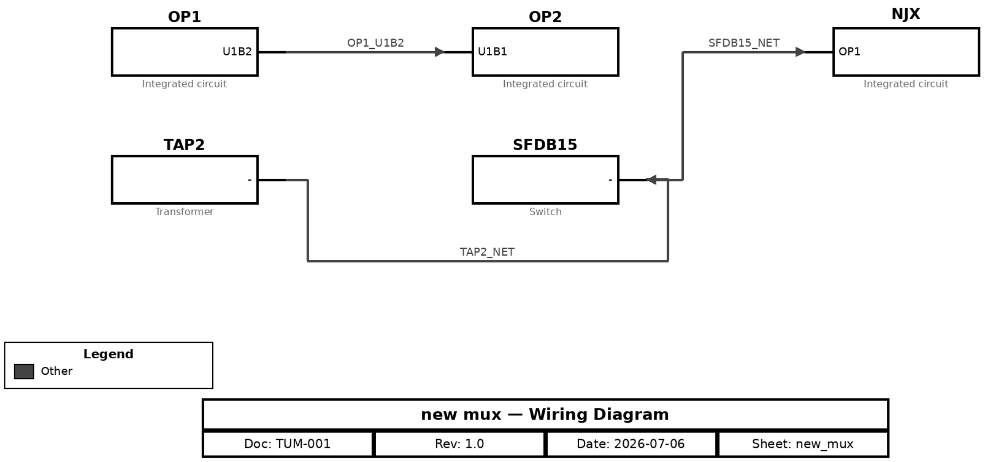
Connected Sub-Module	Via Components	Signals
BB3 SUNRISE	B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR, TAP1	GND, JP6_NET, SFDB15F_NET, TAP1_JP1, TAP1_NET, U1_E, U1_S, U2_E…
GPIO	B2GPIO, B2JX1-CR, B3JX1-CR, B3JX2-CR	B2GPIO_16, B2GPIO_17, B2GPIO_18, GND, U1_E, U1_S, U2_E, U2_S…
JTAG MUX	TAP1	JP6_NET, SFDB15F_NET, TAP1_JP1, TAP1_NET
Scanflex	SFDB15, TAP1	TAP1_NET, TAP2_NET
TB WIRING	B3JX1-CR, B3JX2-CR	JTMUX_1, JTMUX_2
TOP	B3JX1-CR, B3JX2-CR, OP1, XYL-PT2CN	GND, JTMUX_1, JTMUX_2, TCK, TDI, TDO, TMS, VREF
new mux	OP1, OP2, SFDB15	OP1_U1B2, SFDB15_NET, TAP2_NET

Signal List

Signal	Class	Connections
GND	Ground	XYL-PT2CN.8 → OP1.8
VREF	Analog / Sense	XYL-PT2CN.9 → OP1.2
B2JX1-CR_4	Other	B2JX1-CR.4 → B2GPIO.1
B3JX1-CR_10	Other	B3JX1-CR.10 → B2GPIO.7
B3JX1-CR_13	Other	B3JX1-CR.13 → B2GPIO.GND
B3JX1-CR_3	Other	B3JX1-CR.3 → B2GPIO.0
B3JX1-CR_5	Other	B3JX1-CR.5 → B2GPIO.2
B3JX1-CR_6	Other	B3JX1-CR.6 → B2GPIO.3
B3JX1-CR_7	Other	B3JX1-CR.7 → B2GPIO.4
B3JX1-CR_8	Other	B3JX1-CR.8 → B2GPIO.5
B3JX1-CR_9	Other	B3JX1-CR.9 → B2GPIO.6
B3JX2-CR_10	Other	B3JX2-CR.10 → B2GPIO.15
B3JX2-CR_13	Other	B3JX2-CR.13 → B2GPIO.GND
B3JX2-CR_3	Other	B3JX2-CR.3 → B2GPIO.8
B3JX2-CR_4	Other	B3JX2-CR.4 → B2GPIO.9
B3JX2-CR_5	Other	B3JX2-CR.5 → B2GPIO.10
B3JX2-CR_6	Other	B3JX2-CR.6 → B2GPIO.11

Signal	Class	Connections
B3JX2-CR_7	Other	B3JX2-CR.7 → B2GPIO.12
B3JX2-CR_8	Other	B3JX2-CR.8 → B2GPIO.13
B3JX2-CR_9	Other	B3JX2-CR.9 → B2GPIO.14
IP1-U1IP1_NET	Other	IP1-U1IP1.- → IP2-U2IP2.-
IP2-U2IP2_NET	Other	IP2-U2IP2.- → IP3-U3IP3.-
IP3-U3IP3_NET	Other	IP3-U3IP3.- → IP4-U4IP4.-
OP2_U1B2	Other	OP2.U1B2 → TAP1.-
TAP1_NET	Other	TAP1.- → SFDB15.-
TCK	Other	XYL-PT2CN.6 → OP1.4
TDI	Other	XYL-PT2CN.3 → OP1.3
TDO	Other	XYL-PT2CN.4 → OP1.6
TMS	Other	XYL-PT2CN.5 → OP1.5

2.11 new mux



The new mux sub-module interconnects 5 components of the system.

The sub-module integrates 5 components operating together as a single functional stage; the individual parts are listed in the Bill of Materials.

Role in the Overall System

Within the BB3, the new mux sub-module is one of the principal functional stages. It provides supporting functions to the rest of the system. It interfaces with the BB3 SUNRISE sub-module through TAP2 (signals: B3JX2_IP3, TAP2_NET). It interfaces with the JTAG MUX sub-module through TAP2 (signals: B3JX2_IP3, TAP2_NET). It interfaces with the Old Mux sub-module through OP1, OP2, SFDB15 (signals: GND, OP2_U1B2, TAP1_NET, TCK, TDI, TDO, TMS, VREF). It interfaces with the Scanflex sub-module through SFDB15, TAP2 (signals: TAP1_NET, TAP2_NET). It interfaces with the TOP sub-module through OP1 (signals: GND, TCK, TDI, TDO, TMS, VREF).

Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
BB3 SUNRISE	TAP2	B3JX2_IP3, TAP2_NET
JTAG MUX	TAP2	B3JX2_IP3, TAP2_NET
Old Mux	OP1, OP2, SFDB15	GND, OP2_U1B2, TAP1_NET, TCK, TDI, TDO, TMS, VREF
Scanflex	SFDB15, TAP2	TAP1_NET, TAP2_NET
TOP	OP1	GND, TCK, TDI, TDO, TMS, VREF

Signal List

Signal	Class	Connections
OP1_U1B2	Other	OP1.U1B2 → OP2.U1B1
SFDB15_NET	Other	SFDB15.- → NJX.OP1
TAP2_NET	Other	TAP2.- → SFDB15.-

3. Bill of Materials

Curated bill of materials derived from the connectivity data. Entries marked 'TBD' were inferred from connectivity context and require confirmation before procurement.

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
B2GPIO	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B2JX1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B2JX1-CR	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3-JP2	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3-JP4	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3-JP6-JTFPG	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3-XYLPT2	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3JX1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3JX1-CR	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3JX2	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
B3JX2-CR	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3OX1J1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3OX1J7	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3OX2J1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3OX2J7	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3OXJ1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3OXJ2	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3SMPS1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
B3SMPS2	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
BHJP2	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
Brd1	TBD	TBD	TBD	Sensor / Transducer	Inferred from connectivity context — confirm part selection.	1
D3	TBD	TBD	TBD	Diode / LED	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
DB25	TBD	TBD	TBD	Diode / LED	Inferred from connectivity context — confirm part selection.	1
DB37	TBD	TBD	TBD	Diode / LED	Inferred from connectivity context — confirm part selection.	1
DB37B	TBD	TBD	TBD	Diode / LED	Inferred from connectivity context — confirm part selection.	1
DB50	TBD	TBD	TBD	Diode / LED	Inferred from connectivity context — confirm part selection.	1
DB9F	TBD	TBD	TBD	Diode / LED	Inferred from connectivity context — confirm part selection.	1
E4	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
EMU0	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
EMU1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
F10	TBD	TBD	TBD	Fuse / Protective device	Inferred from connectivity context — confirm part selection.	1
F11	TBD	TBD	TBD	Fuse / Protective device	Inferred from connectivity context — confirm part selection.	1
F12	TBD	TBD	TBD	Fuse / Protective device	Inferred from connectivity context — confirm part selection.	1
IP1-U1IP1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
IP2-U2IP2	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
IP3-U3IP3	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
IP4-U4IP4	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
ISDF-10-D	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
J3	TBD	TBD	TBD	Connector	Inferred from connectivity context — confirm part selection.	1
J5	TBD	TBD	TBD	Connector	Inferred from connectivity context — confirm part selection.	1
J6	TBD	TBD	TBD	Connector	Inferred from connectivity context — confirm part selection.	1
JF3	TBD	TBD	TBD	Connector	Inferred from connectivity context — confirm part selection.	1
JP6	TBD	TBD	TBD	Connector	Inferred from connectivity context — confirm part selection.	1
JTMUX	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
K1	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K10	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
K11	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K12	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K13	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K14	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K15	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K16	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K32	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K4	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K41	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K43	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K44	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K45	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
K46	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K47	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K48	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K49	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K5	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K50	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K51	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K52	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K53	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K54	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K55	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K56	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
K57	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K58	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K59	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K6	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K60	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K61	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K62	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K63	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K64	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K65	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K66	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K67	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
K68	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K69	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K7	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K70	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K71	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K72	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K73	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K74	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K75	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K76	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K77	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K78	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
K79	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K8	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K80	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K81	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K82	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K83	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K84	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K85	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K86	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K87	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K88	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
K89	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
K9	TBD	TBD	TBD	Relay / Contactor	Inferred from connectivity context — confirm part selection.	1
LAN9250	TBD	TBD	TBD	Inductor	Inferred from connectivity context — confirm part selection.	1
LM15-23B03			B3SMPS2-ACE	EARTH	EARTH	1
MPP1	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP2	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP3	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP4	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP5	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP6	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP7	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
MPP8	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1
Mux2	TBD	TBD	TBD	Motor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
NJX	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
NJXCR1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
OP1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
OP2	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
OSMUX1			B3OX1J7-2	GND	GND	1
OSMUX2			B3OX2J7-1	12VDC	12VDC	1
R4	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
R6	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
R623			DB25A-14	Resistor		1
RB1	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K19	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K20	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K21	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K22	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
RB1-K23	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K24	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K25	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB1-K26	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K1	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K10	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K11	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K12	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K13	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K14	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K15	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
RB2-K2	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K3	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K4	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K5	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K6	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K7	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K8	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RB2-K9	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RJ	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
RJ45	TBD	TBD	TBD	Resistor	Inferred from connectivity context — confirm part selection.	1
RT125D			B3SMPS1-ACE	EARTH	EARTH	1
SFDB15	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
SFDB15F	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SFDB2	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SW1	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SW4	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SW4.1	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SW4.2	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SW4.3	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
SW4.4	TBD	TBD	TBD	Switch	Inferred from connectivity context — confirm part selection.	1
T11	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
T3	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
T4	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
T7	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
T8	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
TAP1	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
TAP2	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
TAP3	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
TE-42743	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
TP10			DB25A-1	Transformer		1
TP13			DB25A-3	Transformer		1
TP16			DB25A-2	Transformer		1
TP4			DB25A-13	Transformer		1
TP6	TBD	TBD	TBD	Transformer	Inferred from connectivity context — confirm part selection.	1
TP9			DB25A-5	Transformer		1
U1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U1-IP	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U13	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U2	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
U3	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U3-B1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U3-B2	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U3-IP	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U33	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U4	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U4-B1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U4-IP	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U5	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U8	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
U9	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
W12	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
W15	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
W9	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
WG-TOPPR1			U2.2	Integrated circuit		1
WG-TOPPR2			U2.3	Integrated circuit		1
WG-TOPPR3			U2.4	Integrated circuit		1
WG-TOPPR4			U2.5	Integrated circuit		1
WG-TOPPR5			U5.3	Integrated circuit		1
WG-TOPPR6			U8.4	Integrated circuit		1
WG-TOPPR7			U8.5	Integrated circuit		1
WITH	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
XYL-PT2CN	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
XYLPT2	TBD	TBD	TBD	Terminal	Inferred from connectivity context — confirm part selection.	1
cc03r-2830-01	TBD	TBD	TBD	Capacitor	Inferred from connectivity context — confirm part selection.	1
pin1	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1
pin17	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
pin9	TBD	TBD	TBD	Integrated circuit	Inferred from connectivity context — confirm part selection.	1

4. Connectivity Data

BB3

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J6	45	B3OXJ1	1	GND_POWER
U33	AA18	J6	-	SM_RM_GPIO1
J6	-	MPP2	1	SM_RM_GPIO1
MPP2	1	B3OX1J1	9	SM_RM_GPIO1
U33	AB11	J6	-	SM_HS_TXD
J6	-	MPP2	2	SM_HS_TXD
MPP2	2	B3OX1J1	10	SM_HS_TXD
U33	Y11	J6	-	SM_HS_TXC
J6	-	MPP2	3	SM_HS_TXC
MPP2	3	B3OX1J1	11	SM_HS_TXC
U33	AA10	J6	-	SM_HS_RXC
J6	-	MPP2	4	SM_HS_RXC
MPP2	4	B3OX1J1	12	SM_HS_RXC
U33	AB10	J6	-	SM_HS_RXD
J6	-	MPP2	5	SM_HS_RXD
MPP2	5	B3OX1J1	13	SM_HS_RXD
U33	-	K8	-	RMU-SM_UART-Tx
K8	-	J6	-	RMU-SM_UART-Tx
J6	-	MPP2	6	RMU-SM_UART-Tx
MPP2	6	B3OX1J1	14	RMU-SM_UART-Tx
U33	-	K7	-	RMU-SM_UART-Rx
K7	-	J6	-	RMU-SM_UART-Rx
J6	-	MPP2	7	RMU-SM_UART-Rx
MPP2	7	B3OX1J1	15	RMU-SM_UART-Rx
U33	P5	J6	-	SENSE_BLACK1
J6	-	MPP2	8	SENSE_BLACK1
MPP2	8	B3OX1J1	16	SENSE_BLACK1
U33	-	U13	-	PPS_CLK_TO_RMU
U13	-	J6	-	PPS_CLK_TO_RMU
J6	-	MPP3	1	PPS_CLK_TO_RMU
MPP3	1	B3OX1J1	17	PPS_CLK_TO_RMU
U33	AB15	J6	-	PPS_CLK_TO_MB
J6	-	MPP3	2	PPS_CLK_TO_MB
MPP3	2	B3OX1J1	18	PPS_CLK_TO_MB
U33	-	U9	-	SENSE_BLACK2
U9	-	J6	-	SENSE_BLACK2
J6	-	MPP3	3	SENSE_BLACK2
MPP3	3	B3OX1J1	19	SENSE_BLACK2

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
U33	Y15	J6	-	POWER_ON_LBAND_PA2
J6	-	MPP3	4	POWER_ON_LBAND_PA2
MPP3	4	B3OX1J1	20	POWER_ON_LBAND_PA2
U33	Y16	J6	-	POWER_ON_LBAND_PA1
J6	-	MPP3	5	POWER_ON_LBAND_PA1
MPP3	5	B3OX1J1	21	POWER_ON_LBAND_PA1
U33	V2	J6	-	SENSE_BLACK3
J6	-	MPP3	6	SENSE_BLACK3
MPP3	6	B3OX1J1	22	SENSE_BLACK3
U33	-	J3	-	RED2_DLL1
J3	-	J6	-	RED2_DLL1
J6	-	MPP3	7	RED2_DLL1
MPP3	7	B3OX1J1	23	RED2_DLL1
U33	Y10	J6	-	RED2_DLL2
J6	-	MPP3	8	RED2_DLL2
MPP3	8	B3OX1J1	24	RED2_DLL2
U33	G6	J6	-	RED2_DLL3
J6	-	MPP4	1	RED2_DLL3
MPP4	1	B3OX1J1	25	RED2_DLL3
U33	-	E4	-	RED2_DLL4
E4	-	J6	-	RED2_DLL4
J6	-	MPP4	2	RED2_DLL4
MPP4	2	B3OX1J1	26	RED2_DLL4
U33	-	D3	-	RED2_DLL5
D3	-	J6	-	RED2_DLL5
J6	-	MPP4	3	RED2_DLL5
MPP4	3	B3OX1J1	27	RED2_DLL5
U33	AA14	J6	-	POWER_ON_VUHF_PA2
J6	-	MPP4	4	POWER_ON_VUHF_PA2
MPP4	4	B3OX1J1	28	POWER_ON_VUHF_PA2
U33	AB14	J6	-	POWER_ON_VUHF_PA1
J6	-	MPP4	5	POWER_ON_VUHF_PA1
MPP4	5	B3OX1J1	29	POWER_ON_VUHF_PA1
U33	-	W15	-	POWER_ON_BLACK
W15	-	J6	-	POWER_ON_BLACK
J6	-	MPP4	6	POWER_ON_BLACK
MPP4	6	B3OX1J1	30	POWER_ON_BLACK
U33	AA12	J6	-	RED1_DLL1
J6	-	MPP4	7	RED1_DLL1
MPP4	7	B3OX1J1	31	RED1_DLL1
U33	AB12	J6	-	RED1_DLL2
J6	-	MPP4	8	RED1_DLL2

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
MPP4	8	B3OX1J1	32	RED1_DLL2
U33	-	U3	-	RED1_DLL3
U3	-	J6	-	RED1_DLL3
J6	-	MPP5	1	RED1_DLL3
MPP5	1	B3OX2J1	1	RED1_DLL3
U33	-	U1	-	RED1_DLL4
U1	-	J6	-	RED1_DLL4
J6	-	MPP5	2	RED1_DLL4
MPP5	2	B3OX2J1	2	RED1_DLL4
U33	-	T8	-	RED1_DLL5
T8	-	J6	-	RED1_DLL5
J6	-	MPP5	3	RED1_DLL5
MPP5	3	B3OX2J1	3	RED1_DLL5
U33	-	F12	-	RED3_DLL1
F12	-	J6	-	RED3_DLL1
J6	-	MPP5	4	RED3_DLL1
MPP5	4	B3OX2J1	4	RED3_DLL1
U33	-	F11	-	RED3_DLL2
F11	-	J6	-	RED3_DLL2
J6	-	MPP5	5	RED3_DLL2
MPP5	5	B3OX2J1	5	RED3_DLL2
U33	-	F10	-	RED3_DLL3
F10	-	J6	-	RED3_DLL3
J6	-	MPP5	6	RED3_DLL3
MPP5	6	B3OX2J1	6	RED3_DLL3
U33	H10	J6	-	RED3_DLL4
J6	-	MPP5	7	RED3_DLL4
MPP5	7	B3OX2J1	7	RED3_DLL4
U33	G10	J6	-	RED3_DLL5
J6	-	MPP5	8	RED3_DLL5
MPP5	8	DB37B	1	RED3_DLL5
DB37B	1	B3OX2J1	8	RED3_DLL5
U33	AB13	J6	-	MISO_PPS
J6	-	MPP6	1	MISO_PPS
MPP6	1	DB37B	2	MISO_PPS
DB37B	2	B3OX2J1	9	MISO_PPS
U33	Y12	J6	-	MOSI_PPS
J6	-	MPP6	2	MOSI_PPS
MPP6	2	DB37B	3	MOSI_PPS
DB37B	3	B3OX2J1	10	MOSI_PPS
U33	-	W12	-	CLK_PPS
W12	-	J6	-	CLK_PPS

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J6	-	MPP6	3	CLK_PPS
MPP6	3	DB37B	4	CLK_PPS
DB37B	4	B3OX2J1	11	CLK_PPS
U33	-	W9	-	U33_NET
W9	-	J6	-	W9_NET
J6	-	MPP6	4	J6_NET
MPP6	4	DB37B	5	MPP6_4
DB37B	5	B3OX2J1	12	DB37B_5
U33	G11	J6	-	RED4_DLL1
J6	-	MPP6	5	RED4_DLL1
MPP6	5	DB37B	6	RED4_DLL1
DB37B	6	B3OX2J1	13	RED4_DLL1
U33	G12	J6	-	RED4_DLL2
J6	-	MPP6	6	RED4_DLL2
MPP6	6	DB37B	7	RED4_DLL2
DB37B	7	B3OX2J1	14	RED4_DLL2
U33	-	T3	-	RED4_DLL3
T3	-	J6	-	RED4_DLL3
J6	-	MPP6	7	RED4_DLL3
MPP6	7	DB37B	8	RED4_DLL3
DB37B	8	B3OX2J1	15	RED4_DLL3
U33	-	T4	-	RED4_DLL4
T4	-	J6	-	RED4_DLL4
J6	-	MPP6	8	RED4_DLL4
MPP6	8	DB37B	9	RED4_DLL4
DB37B	9	B3OX2J1	16	RED4_DLL4
U33	-	R4	-	SENSE_RED1
R4	-	J6	-	SENSE_RED1
J6	-	MPP7	1	SENSE_RED1
MPP7	1	DB37B	10	SENSE_RED1
DB37B	10	B3OX2J1	17	SENSE_RED1
U33	-	T7	-	SENSE_RED2
T7	-	J6	-	SENSE_RED2
J6	-	MPP7	2	SENSE_RED2
MPP7	2	DB37B	11	SENSE_RED2
DB37B	11	B3OX2J1	18	SENSE_RED2
U33	Y4	J6	-	U33_Y4
J6	-	MPP7	3	J6_NET
MPP7	3	DB37B	12	MPP7_3
DB37B	12	B3OX2J1	19	DB37B_12
U33	-	R6	-	SENSE_RED3
R6	-	J6	-	SENSE_RED3

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J6	-	MPP7	4	SENSE_RED3
MPP7	4	DB37B	13	SENSE_RED3
DB37B	13	B3OX2J1	20	SENSE_RED3
U33	P6	J6	-	SENSE_RED4
J6	-	MPP7	5	SENSE_RED4
MPP7	5	DB37B	14	SENSE_RED4
DB37B	14	B3OX2J1	21	SENSE_RED4
U33	V1	J6	-	SENSE_BLACK4
J6	-	MPP7	6	SENSE_BLACK4
MPP7	6	DB37B	15	SENSE_BLACK4
DB37B	15	B3OX2J1	22	SENSE_BLACK4
U33	-	T11	-	RED4_DLL5
T11	-	J6	-	RED4_DLL5
J6	-	MPP7	7	RED4_DLL5
MPP7	7	DB37B	16	RED4_DLL5
DB37B	16	B3OX2J1	23	RED4_DLL5
U33	-	MPP7	8	U33_NET
U33	-	MPP8	1	U33_NET
U33	-	MPP8	2	U33_NET
U33	-	MPP8	3	U33_NET
U33	-	MPP8	4	U33_NET
U33	-	MPP8	5	U33_NET
U33	-	MPP8	6	U33_NET
U33	-	MPP8	7	U33_NET
U33	-	MPP8	8	U33_NET

BB3 SUNRISE

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
DB25	-	WITH	75	DB25_NET
WITH	75	Brd1	-	WITH_75
DB25	-	Mux2	-	DB25_NET
DB50	-	WITH	75	DB50_NET
DB37	-	DB37B	1	DB37_NET
DB37B	1	B3OXJ2	1	DB37B_1
DB37B	2	B3OXJ2	2	DB37B_2
DB37B	3	B3OXJ2	3	DB37B_3
DB37B	4	B3OXJ2	4	DB37B_4
DB37B	5	B3OXJ2	5	DB37B_5
DB37B	6	B3OXJ2	6	DB37B_6
DB37B	7	B3OXJ2	7	DB37B_7

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
DB37B	8	B3OXJ2	8	DB37B_8
DB37B	9	B3OXJ2	9	DB37B_9
DB37B	10	B3OXJ2	10	DB37B_10
DB37B	11	B3OXJ2	11	DB37B_11
DB37B	12	B3OXJ2	12	DB37B_12
DB37B	13	B3OXJ2	13	DB37B_13
DB37B	14	B3OXJ2	14	DB37B_14
DB37B	15	B3OXJ2	15	DB37B_15
DB37B	16	B3OXJ2	16	DB37B_16
B2GPIO	0	B3JX1-CR	3	U1_S
B2GPIO	1	B2JX1-CR	4	U1_E
B2GPIO	2	B3JX1-CR	5	U2_S
B2GPIO	3	B3JX1-CR	6	U2_E
B2GPIO	4	B3JX1-CR	7	U3_S
B2GPIO	5	B3JX1-CR	8	U3_E
B2GPIO	6	B3JX1-CR	9	U4_S
B2GPIO	7	B3JX1-CR	10	U4_E
B2GPIO	GND	B3JX1-CR	13	GND
B2GPIO	8	B3JX2-CR	3	U1_S
B2GPIO	9	B3JX2-CR	4	U1_E
B2GPIO	10	B3JX2-CR	5	U2_S
B2GPIO	11	B3JX2-CR	6	U2_E
B2GPIO	12	B3JX2-CR	7	U3_S
B2GPIO	13	B3JX2-CR	8	U3_E
B2GPIO	14	B3JX2-CR	9	U4_S
B2GPIO	15	B3JX2-CR	10	U4_E
B2GPIO	GND	B3JX2-CR	13	GND
JP6	-	TAP1	-	JP6_NET
JP6	-	B3-JP6-JTFPG	-	JP6_NET
B3-JP6-JTFPG	-	B2JX1	IP2	B3-JP6-JTFPG_NET
B3-XYLPT2	-	XYLPT2	-	B3-XYLPT2_NET
XYLPT2	-	B3JX1	OP1	XYLPT2_NET
TAP1	-	B3	TAP1	TAP1_NET
B3	TAP1	SFDB15F	-	B3_TAP1
SFDB15F	-	TAP1	JP1	SFDB15F_NET
TAP1	JP1	B3JX1	OP2	TAP1_JP1
BHJP2	A	B3JX2	OP5	BHJP2_A
B3JX2	OP5	B3-JP2	JX1	B3JX2_OP5
B3-JP2	JX1	ISDF-10-D	M	B3-JP2_JX1
ISDF-10-D	M	cc03r-2830-01	g	ISDF-10-D_M
cc03r-2830-01	g	U3	IP	cc03r-2830-01_g
U3	IP	U4	IP	U3_IP

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
U4	IP	B3JX2	IP3	U4_IP
B3JX2	IP3	TAP2	-	B3JX2_IP3
TAP2	-	B3	TAP2	TAP2_NET
B3	TAP2	SFDB2	-	B3_TAP2
SFDB2	-	B3JX2	OP6	SFDB2_NET
U3-B1	3	U3-IP	3	TMS
U3-IP	3	U3-B2	3	TMS
U3-B1	4	U3-IP	4	TDO
U3-IP	4	U3-B2	4	TDO
U3-B1	5	U3-IP	5	TDI
U3-IP	5	U3-B2	5	TDI
U3-B1	6	U3-IP	6	TCK
U3-IP	6	U3-B2	6	TCK
BHJP2	B	B3JX2	OP7	BHJP2_B
B3JX2	OP7	B3-JP4	JX1	B3JX2_OP7
B3-JP4	JX1	B3JX2	IP4	B3-JP4_JX1
U4-B1	2	U4-IP	3	U4-B1_2
EMU0	-	U4-B1	3	EMU0_NET
U4-B1	3	U4-IP	4	U4-B1_3
EMU1	-	U4-B1	4	EMU1_NET
U4-B1	4	U4-IP	5	U4-B1_4
U4-B1	5	U4-IP	6	U4-B1_5

DB CONNECTOR

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
TP10	-	K4	-	TP10_NET
TP16	-	K5	-	TP16_NET
TP13	-	K6	-	TP13_NET
TP6	-	K7	-	TP6_NET
TP9	-	K8	-	TP9_NET
U5	3	B3OXJ2	17	U5_3
B3OXJ2	17	SW4	-	B3OXJ2_17
U8	4	B3OXJ2	18	U8_4
U8	5	B3OXJ2	19	U8_5
U2	2	B3OXJ2	20	U2_2
U2	3	B3OXJ2	21	U2_3
U2	4	B3OXJ2	22	U2_4
U2	5	B3OXJ2	23	U2_5
TP4	-	K9	-	TP4_NET
R623	-	K9	-	R623_NET

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
SW1	1	K10	C	SW1_1
SW1	2	K10	NO	SW1_2
J6	-	K1	-	J6_NET
K1	-	LAN9250	-	K1_NET
J6	-	K32	-	J6_NET
K32	-	RJ	45	K32_NET
J6	2	RJ45	-	B3RJSTL1_2
J6	18	pin1	-	B3RJSTL2_1
pin1	-	pin17	-	B3RJSTL2_1
J6	19	pin9	-	B3RJSTL2_3
pin9	-	J6	-	B3RJSTL2_3
J6	98	RJ45	-	B3RJSTL4_6
J6	11	RJ45	-	B3RJSTL5_3
SW4.1	1	K11	-	SW4.1_1
SW4.1	2	K11	-	SW4.1_2
SW4.2	1	K12	-	SW4.2_1
SW4.2	2	K12	-	SW4.2_2
SW4.3	1	K13	-	SW4.3_1
SW4.3	2	K13	-	SW4.3_2
SW4.4	1	K14	-	SW4.4_1
SW4.4	2	K14	-	SW4.4_2
J6	45	B3OXJ1	1	J6_45
J6	22	B3OXJ1	2	J6_22
J6	24	B3OXJ1	3	J6_24
J6	25	B3OXJ1	4	J6_25
J6	26	B3OXJ1	5	J6_26
J6	27	B3OXJ1	6	J6_27
J6	36	B3OXJ1	7	J6_36
J6	37	B3OXJ1	8	J6_37
J6	39	B3OXJ1	9	J6_39
J6	40	B3OXJ1	10	J6_40
J6	41	B3OXJ1	11	J6_41
J6	42	B3OXJ1	12	J6_42
J6	43	B3OXJ1	13	J6_43
J6	44	B3OXJ1	14	J6_44
J6	46	B3OXJ1	15	J6_46
J6	47	B3OXJ1	16	J6_47
J6	48	B3OXJ1	17	J6_48
J6	49	B3OXJ1	18	J6_49
J6	50	B3OXJ1	19	J6_50
J6	51	B3OXJ1	20	J6_51
J6	53	B3OXJ1	21	J6_53

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J6	54	B3OXJ1	22	J6_54
J6	55	B3OXJ1	23	J6_55
J6	57	B3OXJ1	24	J6_57
J6	58	B3OXJ1	25	J6_58
J6	59	B3OXJ1	26	J6_59
J6	60	B3OXJ1	27	J6_60
J6	61	B3OXJ1	28	J6_61
J6	62	B3OXJ1	29	J6_62
J6	63	B3OXJ1	30	J6_63
J6	64	B3OXJ1	31	J6_64
J6	65	B3OXJ1	32	J6_65
J6	66	DB37B	1	J6_66
DB37B	1	B3OXJ2	1	DB37B_1
J6	77	DB37B	2	J6_77
DB37B	2	B3OXJ2	2	DB37B_2
J6	78	DB37B	3	J6_78
DB37B	3	B3OXJ2	3	DB37B_3
J6	79	DB37B	4	J6_79
DB37B	4	B3OXJ2	4	DB37B_4
J6	80	DB37B	5	J6_80
DB37B	5	B3OXJ2	5	DB37B_5
J6	83	DB37B	6	J6_83
DB37B	6	B3OXJ2	6	DB37B_6
J6	84	DB37B	7	J6_84
DB37B	7	B3OXJ2	7	DB37B_7
J6	85	DB37B	8	J6_85
DB37B	8	B3OXJ2	8	DB37B_8
J6	86	DB37B	9	J6_86
DB37B	9	B3OXJ2	9	DB37B_9
J6	88	DB37B	10	J6_88
DB37B	10	B3OXJ2	10	DB37B_10
J6	89	DB37B	11	J6_89
DB37B	11	B3OXJ2	11	DB37B_11
J6	90	DB37B	12	J6_90
DB37B	12	B3OXJ2	12	DB37B_12
J6	99	DB37B	13	J6_99
DB37B	13	B3OXJ2	13	DB37B_13
J6	100	DB37B	14	J6_100
DB37B	14	B3OXJ2	14	DB37B_14
J6	103	DB37B	15	J6_103
DB37B	15	B3OXJ2	15	DB37B_15
J6	104	DB37B	16	J6_104

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
DB37B	16	B3OXJ2	16	DB37B_16
JP6	4	U1-IP	5	JP6_4
JP6	6	U1-IP	4	JP6_6
JP6	8	U1-IP	6	JP6_8
JP6	10	U1-IP	3	JP6_10

GPIO

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J5	1	DB9F	2	J5_1
J5	2	DB9F	3	J5_2
J5	5	DB9F	5	J5_5
JF3	1	DB9F	3	SCL
JF3	2	DB9F	2	SDA
JF3	3	DB9F	5	GND
B2GPIO	0	B3JX1-CR	3	U1_S
B2GPIO	1	B2JX1-CR	4	U1_E
B2GPIO	2	B3JX1-CR	5	U2_S
B2GPIO	3	B3JX1-CR	6	U2_E
B2GPIO	4	B3JX1-CR	7	U3_S
B2GPIO	5	B3JX1-CR	8	U3_E
B2GPIO	6	B3JX1-CR	9	U4_S
B2GPIO	7	B3JX1-CR	10	U4_E
B2GPIO	GND	B3JX1-CR	13	GND
B2GPIO	8	B3JX2-CR	3	U1_S
B2GPIO	9	B3JX2-CR	4	U1_E
B2GPIO	10	B3JX2-CR	5	U2_S
B2GPIO	11	B3JX2-CR	6	U2_E
B2GPIO	12	B3JX2-CR	7	U3_S
B2GPIO	13	B3JX2-CR	8	U3_E
B2GPIO	14	B3JX2-CR	9	U4_S
B2GPIO	15	B3JX2-CR	10	U4_E
B2GPIO	GND	B3JX2-CR	13	GND
B2GPIO	16	NJXCR1	1	B2GPIO_16
B2GPIO	17	NJXCR1	3	B2GPIO_17
B2GPIO	18	NJXCR1	5	B2GPIO_18
B2GPIO	GND	NJXCR1	8	GND

JTAG MUX

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
JP6	-	TAP1	-	JP6_NET
JP6	-	B3-JP6-JTFPG	-	JP6_NET
B3-JP6-JTFPG	-	B2JX1	IP2	B3-JP6-JTFPG_NET
B3-XYLPT2	-	XYLPT2	-	B3-XYLPT2_NET
XYLPT2	-	B3JX1	OP1	XYLPT2_NET
TAP1	-	B3	TAP1	TAP1_NET
B3	TAP1	SFDB15F	-	B3_TAP1
SFDB15F	-	TAP1	JP1	SFDB15F_NET
TAP1	JP1	B3JX1	OP2	TAP1_JP1
BHJP2	A	B3JX2	OP5	BHJP2_A
B3JX2	OP5	B3-JP2	JX1	B3JX2_OP5
B3-JP2	JX1	ISDF-10-D	M	B3-JP2_JX1
ISDF-10-D	M	cc03r-2830-01	g	ISDF-10-D_M
cc03r-2830-01	g	U3	IP	cc03r-2830-01_g
U3	IP	U4	IP	U3_IP
U4	IP	B3JX2	IP3	U4_IP
B3JX2	IP3	TAP2	-	B3JX2_IP3
TAP2	-	B3	TAP2	TAP2_NET
B3	TAP2	SFDB2	-	B3_TAP2
SFDB2	-	B3JX2	OP6	SFDB2_NET
U3-B1	3	U3-IP	3	TMS
U3-IP	3	U3-B2	3	TMS
U3-B1	4	U3-IP	4	TDO
U3-IP	4	U3-B2	4	TDO
U3-B1	5	U3-IP	5	TDI
U3-IP	5	U3-B2	5	TDI
U3-B1	6	U3-IP	6	TCK
U3-IP	6	U3-B2	6	TCK
BHJP2	B	B3JX2	OP7	BHJP2_B
B3JX2	OP7	B3-JP4	JX1	B3JX2_OP7
B3-JP4	JX1	B3JX2	IP4	B3-JP4_JX1
U4-B1	2	U4-IP	3	U4-B1_2
EMU0	-	U4-B1	3	EMU0_NET
U4-B1	3	U4-IP	4	U4-B1_3
EMU1	-	U4-B1	4	EMU1_NET
U4-B1	4	U4-IP	5	U4-B1_4
U4-B1	5	U4-IP	6	U4-B1_5

Old Mux

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
IP1-U1IP1	-	IP2-U2IP2	-	IP1-U1IP1_NET
IP2-U2IP2	-	IP3-U3IP3	-	IP2-U2IP2_NET
IP3-U3IP3	-	IP4-U4IP4	-	IP3-U3IP3_NET
XYL-PT2CN	9	OP1	2	VREF
XYL-PT2CN	3	OP1	3	TDI
XYL-PT2CN	6	OP1	4	TCK
XYL-PT2CN	5	OP1	5	TMS
XYL-PT2CN	4	OP1	6	TDO
XYL-PT2CN	8	OP1	8	GND
OP2	U1B2	TAP1	-	OP2_U1B2
TAP1	-	SFDB15	-	TAP1_NET
B3JX1-CR	3	B2GPIO	0	B3JX1-CR_3
B2JX1-CR	4	B2GPIO	1	B2JX1-CR_4
B3JX1-CR	5	B2GPIO	2	B3JX1-CR_5
B3JX1-CR	6	B2GPIO	3	B3JX1-CR_6
B3JX1-CR	7	B2GPIO	4	B3JX1-CR_7
B3JX1-CR	8	B2GPIO	5	B3JX1-CR_8
B3JX1-CR	9	B2GPIO	6	B3JX1-CR_9
B3JX1-CR	10	B2GPIO	7	B3JX1-CR_10
B3JX1-CR	13	B2GPIO	GND	B3JX1-CR_13
B3JX2-CR	3	B2GPIO	8	B3JX2-CR_3
B3JX2-CR	4	B2GPIO	9	B3JX2-CR_4
B3JX2-CR	5	B2GPIO	10	B3JX2-CR_5
B3JX2-CR	6	B2GPIO	11	B3JX2-CR_6
B3JX2-CR	7	B2GPIO	12	B3JX2-CR_7
B3JX2-CR	8	B2GPIO	13	B3JX2-CR_8
B3JX2-CR	9	B2GPIO	14	B3JX2-CR_9
B3JX2-CR	10	B2GPIO	15	B3JX2-CR_10
B3JX2-CR	13	B2GPIO	GND	B3JX2-CR_13

Scanflex

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
TAP1	-	TAP2	-	TAP1_NET
TAP2	-	TAP3	-	TAP2_NET
MPP1	4	MPP5	8	MPP1_4
TAP2	-	SFDB15	-	TAP2_NET
TAP1	-	SFDB15	-	TAP1_NET
MPP5	8	DB37B	1	MPP5_8
MPP6	1	DB37B	2	MPP6_1
MPP6	2	DB37B	3	MPP6_2

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
MPP6	3	DB37B	4	MPP6_3
MPP6	4	DB37B	5	MPP6_4
MPP6	5	DB37B	6	MPP6_5
MPP6	6	DB37B	7	MPP6_6
MPP6	7	DB37B	8	MPP6_7
MPP6	8	DB37B	9	MPP6_8
MPP7	1	DB37B	10	MPP7_1
MPP7	2	DB37B	11	MPP7_2
MPP7	3	DB37B	12	MPP7_3
MPP7	4	DB37B	13	MPP7_4
MPP7	5	DB37B	14	MPP7_5
MPP7	6	DB37B	15	MPP7_6
MPP7	7	DB37B	16	MPP7_7

Sheet1

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
U5	3	K9	-	U5_3
U8	4	K10	-	U8_4
U8	5	K11	-	U8_5
U2	2	K12	-	U2_2
U2	3	K13	-	U2_3
U2	4	K14	-	U2_4
U2	5	K15	-	U2_5
TP4	-	K16	-	TP4_NET
R623	-	K41	-	R623_NET
J6	22	K43	-	J6_22
J6	24	K44	-	J6_24
J6	25	K45	-	J6_25
J6	26	K46	-	J6_26
J6	27	K47	-	J6_27
J6	36	K48	-	J6_36
J6	37	K49	-	J6_37
J6	39	K50	-	J6_39
J6	40	K51	-	J6_40
J6	41	K52	-	J6_41
J6	42	K53	-	J6_42
J6	43	K54	-	J6_43
J6	44	K55	-	J6_44
J6	46	K56	-	J6_46
J6	47	K57	-	J6_47

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J6	48	K58	-	J6_48
J6	49	K59	-	J6_49
J6	50	K60	-	J6_50
J6	51	K61	-	J6_51
J6	53	K62	-	J6_53
J6	54	K63	-	J6_54
J6	55	K64	-	J6_55
J6	57	K65	-	J6_57
J6	58	K66	-	J6_58
J6	59	K67	-	J6_59
J6	60	K68	-	J6_60
J6	61	K69	-	J6_61
J6	62	K70	-	J6_62
J6	63	K71	-	J6_63
J6	64	K72	-	J6_64
J6	65	K73	-	J6_65
J6	66	K74	-	J6_66
J6	77	K75	-	J6_77
J6	78	K76	-	J6_78
J6	79	K77	-	J6_79
J6	80	K78	-	J6_80
J6	83	K79	-	J6_83
J6	84	K80	-	J6_84
J6	85	K81	-	J6_85
J6	86	K82	-	J6_86
J6	88	K83	-	J6_88
J6	89	K84	-	J6_89
J6	90	K85	-	J6_90
J6	99	K86	-	J6_99
J6	100	K87	-	J6_100
J6	103	K88	-	J6_103
J6	104	K89	-	J6_104

TB WIRING

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
TE-42743	-	RT125D	-	230V_PH
RT125D	-	B3SMPS1	ACP	230V_PH
LM15-23B03	-	B3SMPS2	ACP	230V_PH
RT125D	-	B3SMPS1	ACN	RT125D_NET
LM15-23B03	-	B3SMPS2	ACN	230V_NU

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
RT125D	-	B3SMPS1	ACE	EARTH
LM15-23B03	-	B3SMPS2	ACE	EARTH
OSMUX1	-	B3OX1J7	1	OSMUX1_NET
OSMUX2	-	B3OX2J7	1	OSMUX2_NET
JTMUX	1	B3JX1-CR	1	JTMUX_1
JTMUX	2	B3JX2-CR	1	JTMUX_2
OSMUX1	-	B3OX1J7	2	GND

TOP

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
WG-TOPPR1	-	U2	2	WG-TOPPR1_NET
WG-TOPPR2	-	U2	3	WG-TOPPR2_NET
WG-TOPPR3	-	U2	4	WG-TOPPR3_NET
WG-TOPPR4	-	U2	5	WG-TOPPR4_NET
WG-TOPPR5	-	U5	3	WG-TOPPR5_NET
WG-TOPPR6	-	U8	4	WG-TOPPR6_NET
WG-TOPPR7	-	U8	5	WG-TOPPR7_NET
TP10	-	K4	-	TP10_NET
TP16	-	K5	-	TP16_NET
TP13	-	K6	-	TP13_NET
TP9	-	K8	-	TP9_NET
TP4	-	K9	-	TP4_NET
R623	-	K9	-	R623_NET
SW1	1	K10	C	SW1_1
SW1	2	K10	NO	SW1_2
XYL-PT2CN	-	OP1	2	VREF
XYL-PT2CN	-	OP1	3	TDI
XYL-PT2CN	-	OP1	4	TCK
XYL-PT2CN	-	OP1	5	TMS
XYL-PT2CN	-	OP1	6	TDO
XYL-PT2CN	-	OP1	8	GND
J6	-	K1	-	J6_NET
J6	-	K32	-	J6_NET
SW4.1	1	K11	-	SW4.1_1
SW4.1	2	K11	-	SW4.1_2
SW4.2	1	K12	-	SW4.2_1
SW4.2	2	K12	-	SW4.2_2
SW4.3	1	K13	-	SW4.3_1
SW4.3	2	K13	-	SW4.3_2
SW4.4	1	K14	-	SW4.4_1

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
SW4.4	2	K14	-	SW4.4_2
MPP5	8	DB37B	1	MPP5_8
MPP6	1	DB37B	2	MPP6_1
MPP6	2	DB37B	3	MPP6_2
MPP6	3	DB37B	4	MPP6_3
MPP6	4	DB37B	5	MPP6_4
MPP6	5	DB37B	6	MPP6_5
MPP6	6	DB37B	7	MPP6_6
MPP6	7	DB37B	8	MPP6_7
MPP6	8	DB37B	9	MPP6_8
MPP7	1	DB37B	10	MPP7_1
MPP7	2	DB37B	11	MPP7_2
MPP7	3	DB37B	12	MPP7_3
MPP7	4	DB37B	13	MPP7_4
MPP7	5	DB37B	14	MPP7_5
MPP7	6	DB37B	15	MPP7_6
MPP7	7	DB37B	16	MPP7_7
RB1	K15	J6	1	ETH_TXP
J6	1	RB1-K19	C	ETH_TXP
RB1	K16	J6	2	EtH_TXN
J6	2	RB1-K20	C	EtH_TXN
RB1	K17	J6	3	ETH_RXP
J6	3	RB1-K21	C	ETH_RXP
RB1	K18	J6	4	ETH_RXN
J6	4	RB1-K22	C	ETH_RXN
RB1	K19	J6	17	ETH_TXP
J6	17	RB1-K23	C	ETH_TXP
RB1	K20	J6	18	EtH_TXN
J6	18	RB1-K24	C	EtH_TXN
RB1	K21	J6	19	ETH_RXP
J6	19	RB1-K25	C	ETH_RXP
RB1	K22	J6	20	ETH_RXN
J6	20	RB1-K26	C	ETH_RXN
RB1	K23	J6	91	ETH_TXP
J6	91	RB2-K1	C	ETH_TXP
RB1	K24	J6	92	EtH_TXN
J6	92	RB2-K2	C	EtH_TXN
RB1	K25	J6	93	ETH_RXP
J6	93	RB2-K3	C	ETH_RXP
RB1	K26	J6	94	ETH_RXN
J6	94	RB2-K4	C	ETH_RXN
RB2	K1	J6	95	ETH_TXP

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
J6	95	RB2-K4	C	ETH_TXP
RB2	K2	J6	96	EtH_TXN
J6	96	RB2-K5	C	EtH_TXN
RB2	K3	J6	97	ETH_RXP
J6	97	RB2-K6	C	ETH_RXP
RB2	K4	J6	98	ETH_RXN
J6	98	RB2-K7	C	ETH_RXN
RB2	K5	J6	9	ETH_TXP
J6	9	RB2-K8	C	ETH_TXP
RB2	K6	J6	10	EtH_TXN
J6	10	RB2-K9	C	EtH_TXN
RB2	K7	J6	11	ETH_RXP
J6	11	RB2-K10	C	ETH_RXP
RB2	K8	J6	12	ETH_RXN
J6	12	RB2-K11	C	ETH_RXN
RB2	K9	J6	13	ETH_TXP
J6	13	RB2-K12	C	ETH_TXP
RB2	K10	J6	14	EtH_TXN
J6	14	RB2-K13	C	EtH_TXN
RB2	K11	J6	15	ETH_RXP
J6	15	RB2-K14	C	ETH_RXP
RB2	K12	J6	16	ETH_RXN
J6	16	RB2-K15	C	ETH_RXN
RB2	K13	J6	30	ETH_TXP-2
RB2	K14	J6	31	EtH_TXN-2
RB2	K15	J6	32	ETH_RXP-2
RB2	K16	J6	33	ETH_RXN-2
RT125D	-	B3SMPS1	ACP	230V_PH
LM15-23B03	-	B3SMPS2	ACP	230V_PH
RT125D	-	B3SMPS1	ACN	RT125D_NET
LM15-23B03	-	B3SMPS2	ACN	230V_NU
RT125D	-	B3SMPS1	ACE	EARTH
LM15-23B03	-	B3SMPS2	ACE	EARTH
OSMUX1	-	B3OX1J7	1	OSMUX1_NET
OSMUX2	-	B3OX2J7	1	OSMUX2_NET
JTMUX	1	B3JX1-CR	1	JTMUX_1
JTMUX	2	B3JX2-CR	1	JTMUX_2
OSMUX1	-	B3OX1J7	2	GND

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Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
OP1	U1B2	OP2	U1B1	OP1_U1B2
TAP2	-	SFDB15	-	TAP2_NET
SFDB15	-	NJX	OP1	SFDB15_NET